

Foundations of Chemistry

- ❑ **Chemistry** is the study of **elements** in the **Universe**, encompassing their **nature**, occurrence, **physical and chemical** properties, compounds, **reactivity**, uses, and applications.
- ❑ The word "**chemie**" originates from **Khem**, a synonym for **Egypt**, reflecting the Western belief that all knowledge originated from Egypt.
- ❑ In Sanskrit, the language of ancient India, Chemistry was known as **Rasayana Sastra**, **Rasa Sastra**, **Rasa Vada**, or **Rasa Prakriya**, as evidenced by literary works and archaeological excavations in India.
- ❑ The evolution of knowledge over time across different **nations** and **civilizations** can be understood only through available evidence and literature.

Acharya Prafulla Chandra Ray

- ❖ **Acharya Prafulla Chandra Ray (1861–1944)**-Father of Indian Chemistry.
- ❖ Conducted notable studies on **inorganic and industrial chemistry**.
- ❖ Pioneered research in **nitrites and mercurous compounds**.
- ❖ **Founded India's first pharmaceutical company**, *Bengal Chemicals & Pharmaceuticals*.
- ❖ Fellow of the **Royal Society of Chemistry**; honored with CIE (Companion of the Indian Empire) in 1912.
- ❖ Advocated for swadeshi movement, science education, and self-reliance.



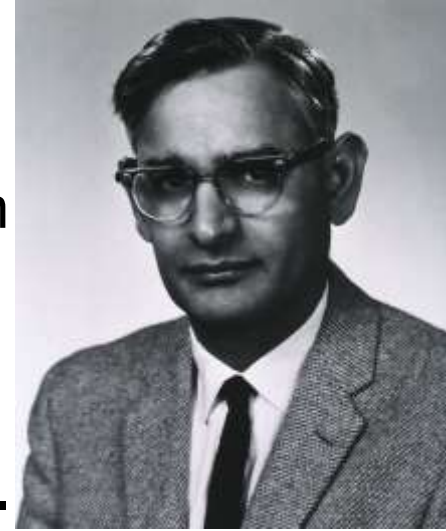
Chintamani Nagesa Ramachandra Rao

- ❖ A pioneering Indian scientist in **solid-state and materials chemistry**.
- ❖ Made transformative contributions to **transition-metal oxides, superconductivity, and nanomaterials**.
- ❖ Published **>1,600 research papers** and **more than 50 books**.
- ❖ Honored with the **Bharat Ratna** in 2014, jointly with Sachin Tendulkar.



Har Gobind Khorana

- ❖ Pioneered work in **decoding the genetic code**.
- ❖ Synthesized the **First artificial genes**, establishing a foundation for modern genetic engineering and biotechnology.
- ❖ Shared the **1968 Nobel Prize in Physiology or Medicine** with **Marshall W. Nirenberg** and **Robert W. Holley** for interpreting the genetic code.
- ❖ **Developed techniques for synthesizing oligonucleotides**, tools that are now indispensable in biotechnology, genetics, and medical research.
- ❖ Contributed significantly to understanding the role of nucleotides in **photosynthesis and membrane function**.



Shanti Swarup Bhatnagar



- ❖ Father of Research Laboratories in India, pioneer of modern scientific infrastructure.
- ❖ Key founder of **CSIR (1942)** and its **first Director-General**.
- ❖ Contributed to **applied physical chemistry**, developing industrial processes using locally available resources.
- ❖ Advanced **magneto-chemistry** and co-developed the Bhatnagar-Mathur Magnetic Interference Balance.
- ❖ The Shanti Swarup Bhatnagar Prize for Science and Technology honors outstanding Indian scientists.

Raghunath Anant Mashelkar



- ❖ Pioneered polymer science innovations, especially in **non-Newtonian fluids and smart materials**.
- ❖ Advocated for intellectual property rights, **shaping policy and fostering innovation-driven growth in India**.
- ❖ **Led CSIR (1995–2006)**, transforming it into a globally competitive, impact-driven research network.
- ❖ Championed affordable innovation through “**Gandhian Engineering**” - more from less for more people.
- ❖ Key contributor to developing **India's patent** system and strengthening **R&D** across sectors.

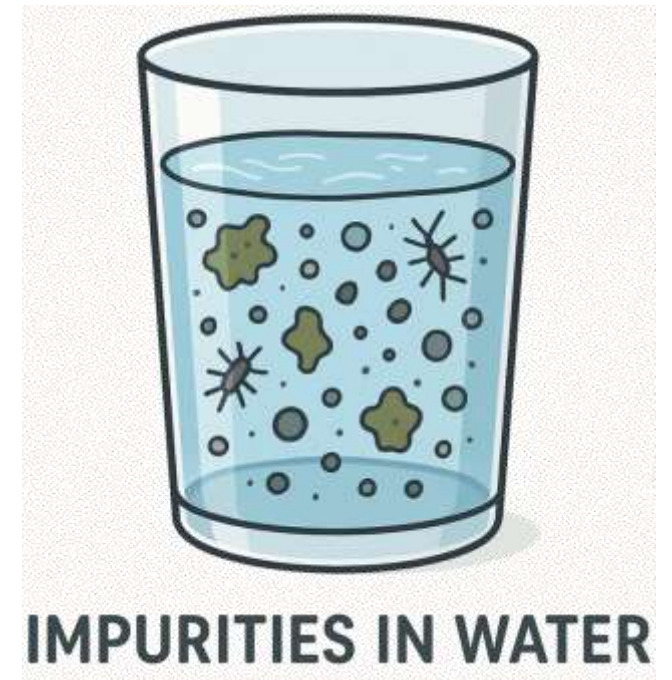
Venkatraman Ramakrishnan

- ❖ Revealed the **ribosome's structure** and mechanism that **translates genetic information into proteins**.
- ❖ Shared the **2009 Nobel Prize** in Chemistry for studies on the structure and function of the ribosome.
- ❖ Advanced **X-ray crystallography and cryo-EM** for studying complex biological molecules.
- ❖ President of the **Royal Society (2015–2020)**, promoting global scientific collaboration and public engagement.

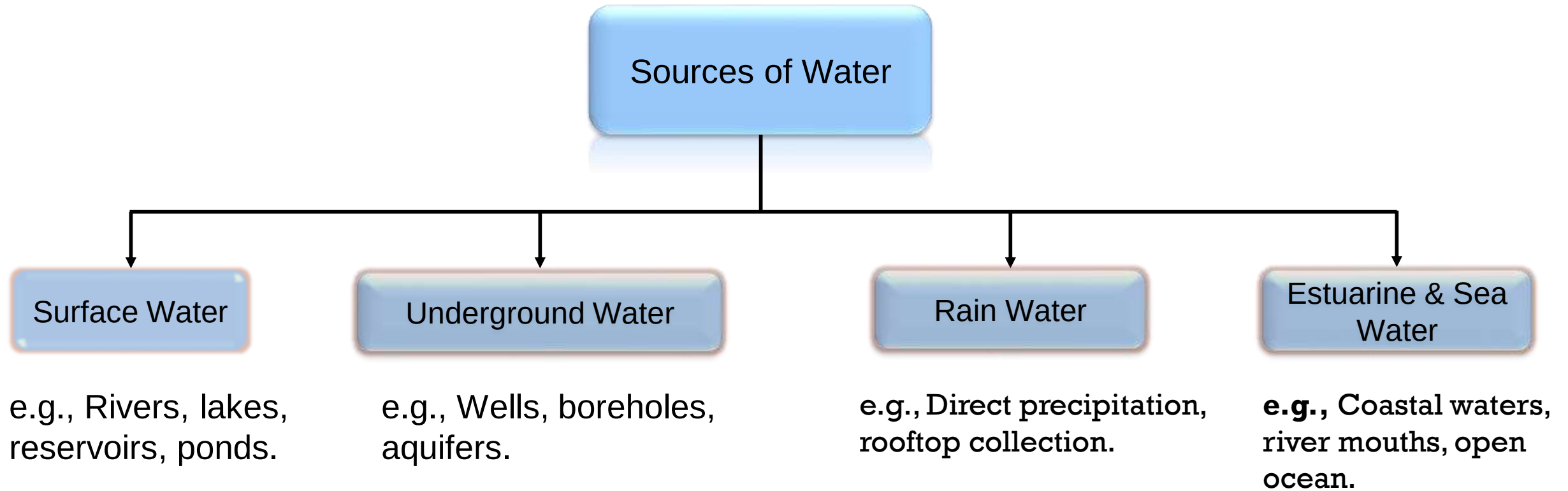


WATER TECHNOLOGY

- ❖ The process of making **water safe**, **clean**, and suitable for its intended purpose, whether for **drinking**, **industrial** use, **agriculture**, or **environmental restoration**.
- ❖ **Main Aim:** To remove or reduce **physical**, **chemical**, and **biological contaminants**, and to adjust water properties such as **pH**, **hardness**, **taste**, and **odor**.
- ❖ **Why Needed?** Raw water from nature often contains **suspended solids**, **dissolved minerals**, **organic matter**, or **microorganisms** that may be harmful or unpleasant.



Introduction of water treatment



Sources of Water: Water for treatment plants originates from various **natural sources**, each with distinct characteristics and quality concerns that influence the treatment process.

a) Surface Water: Water found on the earth's surface-rivers, lakes, ponds, and reservoirs.

- ✓ **Advantages:** Usually abundant, easier to access.
- ✓ **Challenges:** Prone to **suspended solids**, **microorganisms**, and contamination from **industrial**, **agricultural**, or **urban runoff**.

b) Underground Water: Water found beneath the earth's surface, stored in aquifers.

Types:

- ✓ **Shallow wells** – closer to the surface, more prone to contamination.
- ✓ **Deep wells / Tube wells** – draw from deeper aquifers, generally cleaner but may have high mineral content.

Advantages: Less exposure to surface contamination, often free from pathogens.

Challenges: May be hard or contain high salts, iron, fluoride, or arsenic.

c) Rainwater: Water obtained directly from rainfall.

- ✓ **Advantages:** Naturally soft, low mineral content.
- ✓ **Challenges:** Collection depends on rainfall patterns; risk of contamination from roofing materials, air pollutants, and storage tanks.
- ✓ **Typical use:** Supplementary water source in rural areas or water-scarce regions.

d) Estuarine Water: Found in estuaries where freshwater from rivers mixes with seawater, forming **brackish water** with variable salinity influenced by **tides, river flow, and seasons**; typically nutrient-rich but susceptible to **pollution** from nearby populated or industrial areas.

Advantages

- ✓ Nutrient-rich, supporting diverse and productive ecosystems.
- ✓ Acts as a natural buffer, filtering pollutants from rivers before reaching the ocean.

Challenges

- ✓ Variable salinity and quality make treatment more complex.
- ✓ Highly susceptible to pollution from urban, agricultural, and industrial runoff.

Introduction of water treatment

Sea Water: Originates from open oceans and seas, with high salinity (~35 ppt) and containing dissolved salts, minerals, and gases; generally stable in composition, though temperature, salinity, and dissolved oxygen can vary with location and depth.

Advantages

- ✓ Abundant and globally available resource.
- ✓ Relatively stable chemical composition.

Challenges

- ✓ High salinity requires energy-intensive desalination for most uses.
- ✓ Potential contamination from oil spills, microplastics, and industrial waste.

Dissolved Impurities in Water

- ✓ **Inorganic salts** – e.g., calcium, magnesium, sodium, potassium chlorides, sulfates, nitrates.
- ✓ **Dissolved gases** – e.g., oxygen (O_2), carbon dioxide (CO_2), hydrogen sulfide (H_2S).
- ✓ **Minerals** – Naturally occurring from soil and rock dissolution.
- ✓ **Organic compounds** – Natural organics (humic substances) or synthetic chemicals (pesticides, industrial effluents).
- ✓ **Heavy metals** – e.g., lead, mercury, arsenic, cadmium (often from industrial pollution).

Type of impurity present in water

Suspended Impurities in Water

Solid particles that **do not dissolve in water** and remain **floating or dispersed**.

Sources: Soil erosion, surface runoff, construction activities, industrial discharge, sewage.

e.g., Sand, clay, silt, algae, plankton, industrial waste particles.

Bacterial Impurities in Water

Presence of pathogenic (disease-causing) or non-pathogenic microorganisms in water.

e.g., *Escherichia coli* (E. coli), *Salmonella*, *Vibrio cholerae*, *Shigella*, coliform bacteria.

Sources:

- ✓ Sewage and wastewater discharge
- ✓ Animal waste runoff
- ✓ Contaminated surface water and shallow groundwater
- ✓ Faulty septic systems

Effect of Impurities on Water Quality Parameters

Color: Impurities such as dissolved organic matter, metals (iron, manganese), or industrial waste can cause yellow, brown, or reddish tints.

Taste: Dissolved salts, metals, and biological growth can impart salty, metallic, or bitter taste.

Odor: Decaying organic matter, algae, or bacterial contamination can cause musty, rotten egg, or earthy smells.

Turbidity: Suspended particles like clay, silt, and microorganisms make water cloudy, reducing transparency.

Dissolved Hardness: Calcium and magnesium salts increase water hardness, causing scaling in pipes and appliances.

Alkalinity: The Presence of bicarbonates, carbonates, and hydroxides can make water more alkaline, affecting taste and pH balance.

Total Dissolved Solids (TDS): High levels from salts, minerals, and metals can affect taste, cause scaling, and reduce suitability for drinking or industrial use.

Common stages in water treatment

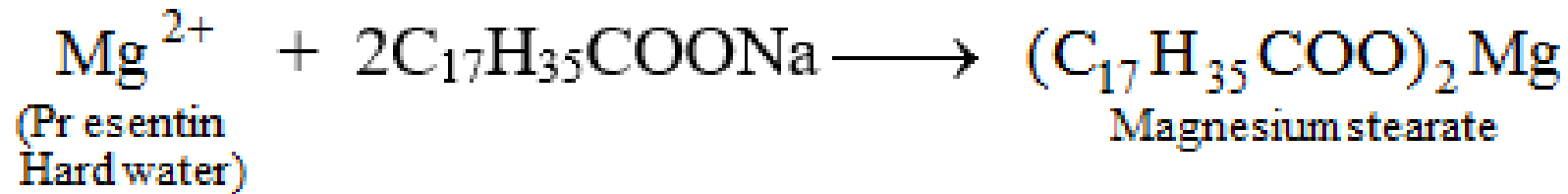
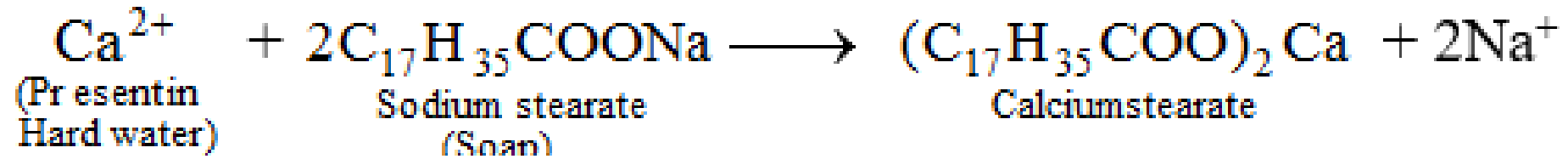
1. **Screening** – removes large debris.
2. **Coagulation & Flocculation** – clumps fine particles together.
3. **Sedimentation** – lets particles settle.
4. **Filtration** – removes remaining suspended matter.
5. **Disinfection** – kills pathogens (e.g., with chlorine, UV light, ozone).

Hardness of Water

The water which **do not produce lather** or produce **very little lather** with soap are called Hard water.

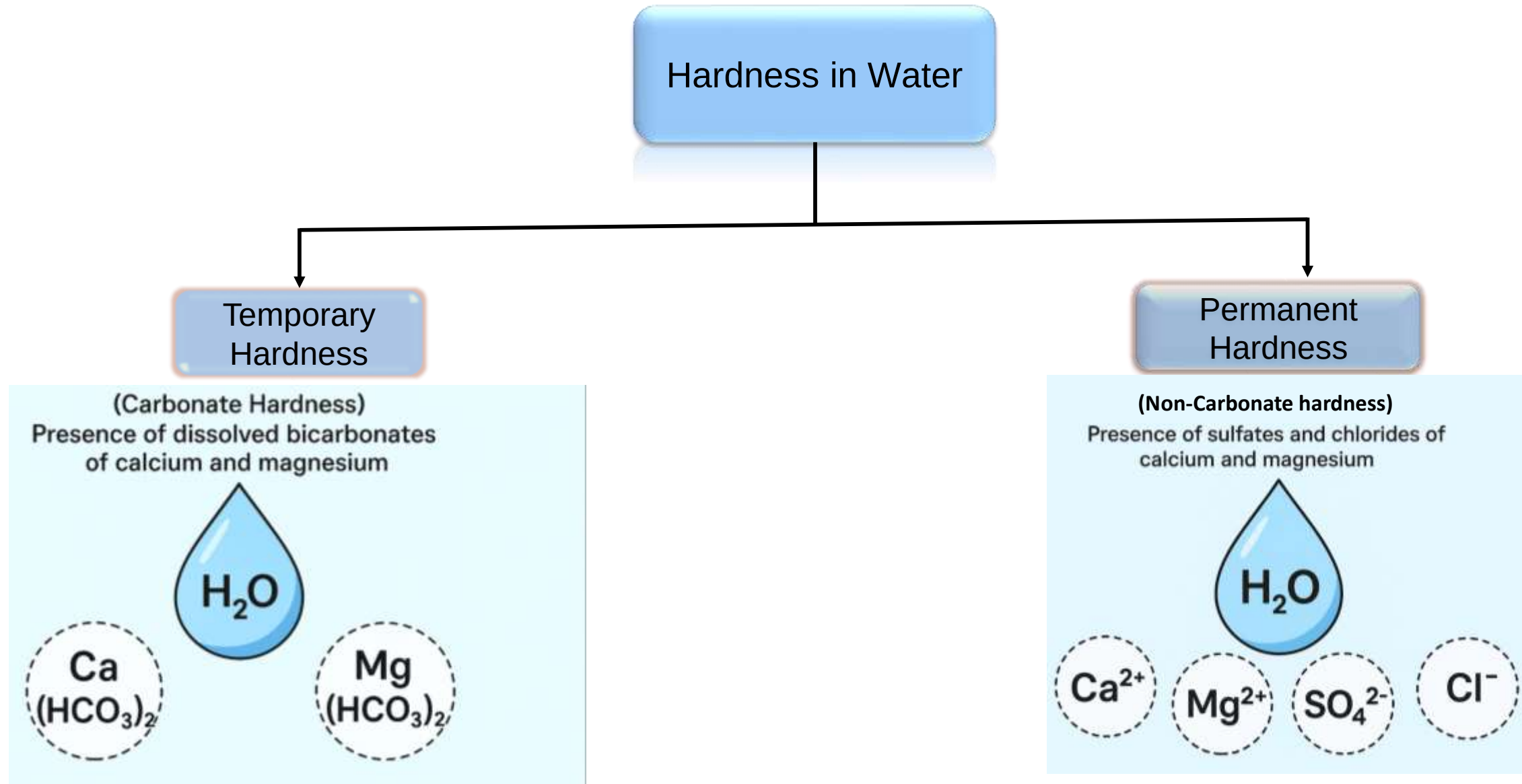
Hardness

- Hardness was originally defined as the **“SOAP CONSUMING CAPACITY”** of a water sample
- It is mainly due to the presence of **Ca & Mg ions**.
- These ions react with the **sodium salts of long-chain fatty acids** present in the soap.
- Formation of **insoluble scums** will occur. Which do not possess any detergent value.



- Other metal ions such as Fe^{2+} , Mn^{2+} , and Al^{3+} also react with soap in the same way as calcium and magnesium, thereby contributing to hardness.
- However, these ions are usually found only in small amounts in natural waters like rainwater and snowfall.

Types of Hardness in Water

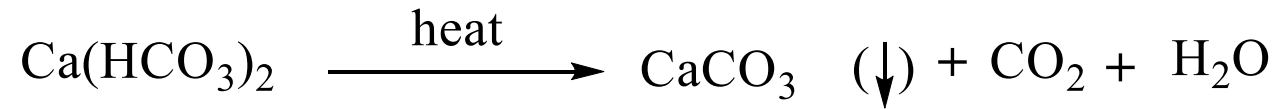


1. Temporary Hardness (Carbonate Hardness)

Cause: Presence of dissolved **bicarbonates** of calcium and magnesium $\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$.

Nature: Can be removed by simple heating (**boiling**).

Reaction During Boiling:



Insoluble calcium carbonate (CaCO_3) precipitates out, **reducing hardness**.

Removal Methods: **Boiling** (precipitates bicarbonates), **Lime softening** (adding $\text{Ca}(\text{OH})_2$ to precipitate carbonates).

e.g., Hardness in river or lake water due to **limestone dissolution**.

2. Permanent Hardness (Non-Carbonate Hardness)

Cause: Presence of sulfates and chlorides of calcium and magnesium CaSO_4 , MgSO_4 , CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4 , CaCl_2 , MgCl_2 .

Nature: Cannot be removed by boiling.

Removal Methods:

- ✓ Ion-exchange method (water softeners)
- ✓ Zeolite method (exchanging Ca^{2+} , Mg^{2+} with Na^+ ions)
- ✓ Reverse Osmosis (RO) filtration

e.g., Groundwater in coastal or mineral-rich areas.

Sources of Hardness in Water

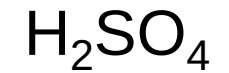
Temporary hardness



Permanent hardness



Non-hardness causing



Org. mat

Unit of Hardness in Water

Hardness is a measure of the **concentration of salts of Ca & Mg salts in terms of CaCO_3 equivalent** and can be expressed in several different units.

1. Part per million (ppm):

1 ppm = 1 part of CaCO_3 equivalent hardness of 10^6 parts of water.

2. Mg/L: 1mg/L = 1 mg of CaCO_3 equivalent per 1 liter of water (1 liter of water = 10^6 mg)

OR

1 mg/L = 1 part of CaCO_3 equivalent hardness of 10^6 parts of water.

3. Degree Frenc ($^{\circ}\text{F}$)

1°F = 1 part of CaCO_3 equivalent hardness of 10^5 parts of water. (1°F = 10 ppm hardness)

4. Degree Clark ($^{\circ}\text{Cl}$)

1°Cl = 1 part of CaCO_3 equivalent hardness of 7×10^4 parts of water.

Relation between Units:

$$1 \text{ ppm} = 1 \text{ mg/L} = 0.1 \text{ }^{\circ}\text{F} = 0.07 \text{ }^{\circ}\text{C}$$

Degree of Hardness

- Degree of hardness is defined as the amount of CaCO_3 equivalent hardness present in a definite number of parts of water, depending upon the units in which hardness is expressed.
- Hardness is never due to CaCO_3 because calcium carbonate is **insoluble in water**
- Expression of hardness in terms of **CaCO_3 equivalent for convenience**.
- Its **molecular weight (100) and equivalent weight (50)** make calculations easy.

Equivalent of CaCO_3 for Hardness-Causing Substances

$$\text{CaCO}_3 \text{ Eq.} = \text{Given wt. of impurity} * \frac{\text{Eq. wt. of CaCO}_3}{\text{Eq. wt. of impurity}}$$

$$\text{Eq. wt.} = \frac{\text{Molar mass}}{n\text{-factor}}$$

For **calcium carbonate (CaCO_3)**:

Molar mass of $\text{CaCO}_3 = 40 + 12 + 3 \times 16 = 100 \text{ g mol}^{-1}$

(since it produces Ca^{2+} and CO_3^{2-})

So, $n\text{-factor} = 2$

$$\text{Eq. wt. of CO}_3 = \frac{100}{2} = 50$$

Numerical Problems

Example 1: Determination of Hardness from Dissolved Salts

A water sample is found to contain the following dissolved salts:

16.2 mg/L of calcium bicarbonate, $\text{Ca}(\text{HCO}_3)_2$

7.3 mg/L of magnesium bicarbonate, $\text{Mg}(\text{HCO}_3)_2$

9.5 mg/L of magnesium chloride, MgCl_2

13.6 mg/L of calcium sulfate, CaSO_4

☞ Calculate the total hardness of the given water sample.

Solution:

Step 1: Convert to CaCO_3 equivalent

$$\text{CaCO}_3 \text{ Eq.} = \text{Given wt. of impurity} * \frac{\text{Eq. wt. of CaCO}_3}{\text{Eq. wt. of impurity}}$$

a) Calcium bicarbonate, $\text{Ca}(\text{HCO}_3)_2$

- Molar mass = $40 + 2 \times (1 + 12 + 3 \times 16) = 40 + 2 \times 61 = 162 \text{ g/mol}$
- One $\text{Ca}^{2+} \rightarrow$ valency = Equivalent weight = $162 / 2 = 81 \text{ g/eq}$
- Hardness:

$$\frac{16.2 \times 50}{81} = 10 \text{ mg/L as } \text{CaCO}_3$$

OR

$$\frac{16.2 \times 100}{162} = 10 \text{ mg/L as } \text{CaCO}_3$$

b) Magnesium bicarbonate, $\text{Mg}(\text{HCO}_3)_2$

- Molar mass = $24.3 + 2 \times 61 = 146.3 \text{ g/mol}$

- Hardness:

$$\frac{7.3 \times 100}{146} = 5 \text{ mg/L as } \text{CaCO}_3$$

c) Magnesium chloride, MgCl_2

- Molar mass = $24.3 + 2 \times 35.5 = 95.3 \text{ g/mol}$

- Hardness:

$$\frac{9.5 \times 100}{95} = 10 \text{ mg/L as } \text{CaCO}_3$$

d) Calcium sulfate, CaSO_4

- Molar mass = $40 + 32 + 4 \times 16 = 136 \text{ g/mol}$
- Hardness:

$$\frac{13.6 \times 100}{136} = 10 \text{ mg/L as } \text{CaCO}_3$$

Step 2: Calculate temporary and permanent hardness:

Temporary Hardness: (from bicarbonates): 10 mg/L (from $\text{Ca}(\text{HCO}_3)_2$) + 5 mg/L (from $\text{Mg}(\text{HCO}_3)_2$) = 15 mg/L or 15 ppm

Permanent Hardness: (from chlorides and sulfates): 10 mg/L (from MgCl_2) + 10 mg/L (from CaSO_4) = 20 mg/L or 20 ppm

Step 3: Total hardness

Now add all contributions:

$$10 \text{ mg/L} + 5 \text{ mg/L} + 10 \text{ mg/L} + 10 \text{ mg/L} = 35 \text{ mg/L as } \text{CaCO}_3$$

Example 2: Calculate the **temporary hardness** and the **total hardness** of a water sample that contains the following dissolved salts: $\text{Mg}(\text{HCO}_3)_2 = 73 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 162 \text{ mg/L}$, $\text{MgCl}_2 = 95 \text{ mg/L}$, $\text{CaSO}_4 = 136 \text{ mg/L}$, $\text{NaCl} = 94 \text{ mg/L}$.

Solution: calculation of CaCO_3 equivalents:

Constituent	Multiplication factor	CaCO_3 equivalent
$\text{Mg}(\text{HCO}_3)_2 = 73 \text{ mg/L}$	$73 \times 100 / 146$	50 mg/L
$\text{Ca}(\text{HCO}_3)_2 = 162 \text{ mg/L}$	$162 \times 100 / 162$	100 mg/L
$\text{MgCl}_2 = 95 \text{ mg/L}$	$95 \times 100 / 95$	100 mg/L
$\text{CaSO}_4 = 136 \text{ mg/L}$	$136 \times 100 / 136$	100 mg/L
$\text{NaCl} = 94 \text{ mg/L}$	Not produce hardness	-

∴ **Temporary hardness** of water due to $[\text{Mg}(\text{HCO}_3)_2 \text{ and } \text{Ca}(\text{HCO}_3)_2]$

$$= 100 + 50$$

$$= 150 \text{ mg/L or ppm.}$$

∴ **Permanent Hardness** of water is due to $[\text{MgCl}_2 \text{ and } \text{CaSO}_4]$

$$= 100 + 100$$

$$= 200 \text{ mg/L or ppm}$$

∴ **Total hardness of water is** $= 50 + 100 + 100 + 100 = 350 \text{ mg/L or ppm.}$

Estimation of hardness using EDTA

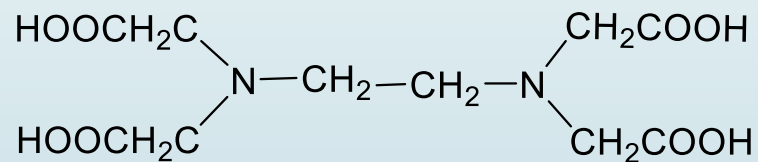
► **Aim:** To estimate the amount of hardness present in the given water sample.

► **Chemicals Required:**

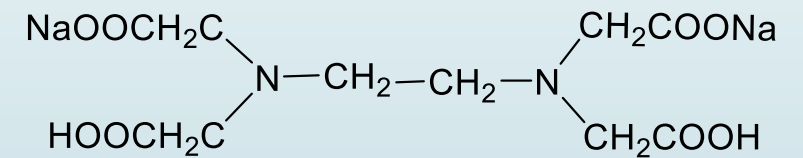
- Standard hard water
- EDTA solution
- EBT Indicator
- Ammonia buffer

Principle

- EDTA is a powerful **chelating agent**, a molecule that can securely bind to metal ions (like Ca^{2+} , Mg^{2+} , Fe^{3+} , Pb^{2+}) using multiple atoms in its structure.
- It is **insoluble in water**.

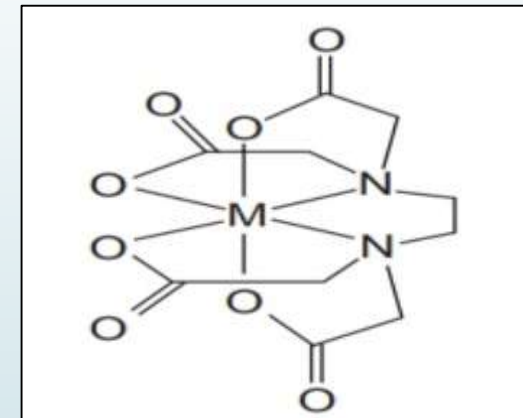


Insoluble in water



Na_2EDTA
soluble in water

- Formation of complex with metal ions such as Mg^{2+} , Ca^{2+} .



Eriochrome Black-T (EBT)

PH	5.5	7-11	Above 11
Color	Red	Blue	Yellowish-orange

- The indicator is rarely applied in titrations of solutions more acidic than pH = 6.5.
- To maintain the pH constant (i.e.10) a **buffer mixture** ($\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$) is added.
- In the pH range 7-11 the addition of metallic salts produces a brilliant change in color **from blue to wine red**.
- This color change can be observed with the ions of Mg, Mn, Zn, Cd, Hg, Pb, Cu, Al, Fe, Ti, Co, Ni, and the Pt metals.

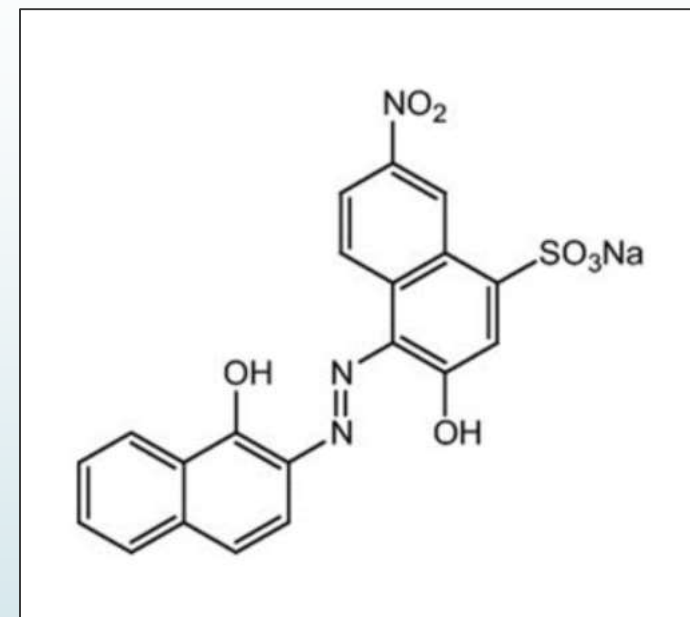


Fig. Structure of EBT

➡ Procedure:

1) Preparation of Standard Hard water

- Dissolve 1 g CaCO_3 in a minimum quantity of dil. HCl



- Then evaporate the solution to dryness on a water bath. (to expel excess of acid & CO_2)
- Dissolve residue in distilled water to make 1-liter solution.
- Each mL of this solution thus contains 1 mg of CaCO_3 eq. hardness.

2) Preparation of EDTA solution:

Dissolve 4 gm of pure EDTA Crystal + 0.1 gm MgCl_2 in 1 litre DW.

3) Preparation of the indicator:

Dissolve 0.5 gm of EBT indicator in 100 ml of alcohol.

4) Preparation of Buffer Solution

Dissolve 67.5 g of NH_4Cl + 570 ml of conc. NH_4OH solution in 1 liter DW.

5) Standardization of EDTA Solution

- Pipette out 50 mL of std. Hard water (**Known**) in a conical flask.
- Add 10-15 mL of buffer solution & 4 to 5 Drops of indicator.
- Titrate with EDTA solution till the wine-red colour changes to blue
- Let volume used by V_1 mL

6) Titration of Unknown hard water

Titrate 50 mL of water sample just in step (5). Let the volume used be V_2 ml.

7) Titration of permanent Hardness

- Take 250 mL of the water sample.
- Boil it till the volume is reduced to about 50 mL
- Filter & wash it
- Collect it
- Finally, make up the volume to 250 ml with DW.
- Then, Titrate 50mL of the boiled water sample just as in step (5). Let volume used by V_3 ml.

► Calculations:

8

1) Standardization of EDTA solution:

Since 1000 mL of std. water contains = 1000 mg of CaCO_3

Therefore, 1 mL of std. water contains = 1mg of CaCO_3 (1)

Also, 50 mL of std. hard water consists = V1 mL of EDTA

V1 mL of EDTA = 50 mg of CaCO_3 equivalent

1 mL of EDTA = $50/V1$ mg of CaCO_3 equivalent (2)

2) Determination of the total Hardness of water

Since 50 mL of hard water = V2 mL of EDTA

Therefore, 1 mL of hard water = $V2/50$ mL of EDTA

Hence, 1000 mL of hard water = $(V2/50) \times 1000$ mL of EDTA

from equation 2

1mL of EDTA = $50/V_1$ mg of CaCO_3 equivalent

Therefore, **Total hardness** = $(V_2/50) * 1000 * (50/V_1)$ mg of CaCO_3 equivalent
= $(V_2/V_1) * 1000$ mg of CaCO_3 equivalent
= **$(V_2/V_1) * 1000$ mg/L or ppm**

3) Determination of Permanent Hardness:

Since 50 mL boiled water sample = V_3 mL EDTA

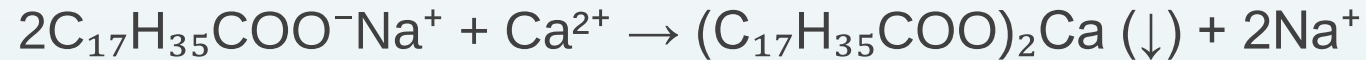
Therefore, 1000 mL of boiled water sample = $(V_3/50) * 1000$ mL of EDTA

from eq. 2

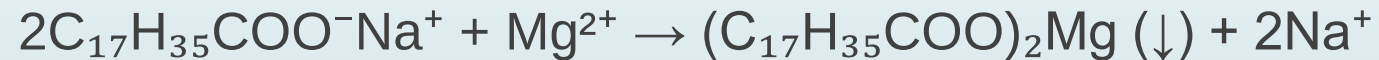
Permanent Hardness = $(V_3/50) * 1000 * (50/V_1)$ mg/L
= $(V_3/V_1) * 1000$ mg/L

Estimation of hardness using the soap solution method

- Soluble soaps consist of **sodium or potassium salts of higher fatty acids**, such as oleic acid, stearic acid, and palmitic acid.



Sod. Stearate(soluble soap) Cal. Stearate(insoluble soap)



(soluble soap)

(insoluble soap)

- These soaps **give lather** with hard water only **after a sufficient quantity of the soap is added to precipitate** all the hardness-causing metal ions present in water.
- Thus, after precipitation, the total hardness can be determined by titrating against a standard soap solution in alcohol.

Lather factor: Volume of soap solution required to form lather with distilled water.

$$\text{Lather factor} = V_{\text{lather}} / V_{\text{solution}}$$

V_{lather} = total volume of foam (mL) generated,

V_{solution} = volume of soap solution (mL) used to generate the foam.

Procedure

1. Preparation of Std. Hard water

- Dissolve **1 g CaCO_3** in a **minimum quantity of dil. HCl**
- Then evaporate the solution to dryness in a water bath. (to expel excess of acid & CO_2)
- Dissolve residue in DW to make a 1-liter solution.
- **Each mL of this solution** thus contains **1 mg of CaCO_3 equiv. hardness**

2. Preparation of Soap solution

- Dissolve 100 g of pure dry castile soap in 800 mL of alcohol & 200mL of DW.

3. Standardization of Soap solution

- Take 50 ml of Std. Hard water (**Known**) in a narrow-mouth 250 mL glass stoppered bottle.
- Add it to the soap solution dropwise. Shake the bottle and let it stand for a short time.
- See if a lather is formed or not.
- Note the time for which lather persists.
- Conti. Titration until a lather forms after vigorous shaking lasts for 2 minutes.
- Let the volume of soap solution used be V_1 mL.

4. Lather factor of distilled water

- Titrate 50 mL of DW against soap solution.
- Till lather persists for 2 minutes.
- Let the volume used be V mL.
- **This volume is to be subtracted from all values (Standardization of soap solution, or Total Hardness, or Permanent hardness)**

5. Total Hardness

- Repeat above titration
- Taking 50ml of water sample (Unknown)
- Let the titre value of soap solution be V_2 mL

6. Permanent hardness

- Take 250 ml of the water sample.
- Boil it till the volume is reduced to about 50 mL
- Filter & wash it
- Collect it
- Finally, make up the volume to 250 mL with DW.
- Then, Titrate 50ml of the boiled water sample against the soap solution.
- Let the volume used by V_3 mL.

► Calculations

1) Standardization of Soap solution:

Since 1000 mL of std. water contains = 1000 mg of CaCO_3

Therefore, 1 mL of std. water contains = 1 mg of CaCO_3 (1)

Also, 50 mL of std. hard water consists = (V1-V) mL of Soap Solution

(V1-V) mL of Soap Solution = 50 mg of CaCO_3 equivalent

1 mL of Soap solution = $50/(V1-V)$ mg of CaCO_3 equivalent (2)

2) Determination of the total Hardness of water

Since 50 mL of hard water = (V2-V) mL of Soap Solution

Therefore, 1 mL of hard water = (V2-V)/50 mL of Soap Solution

Hence, 1000 mL of hard water = (V2-V)/50*1000 mL of Soap Solution

from equation 2

1 mL of Soap solution = $50 / (V1 - V)$ mg of CaCO_3 equivalent

Therefore, **Total hardness** = $(V2 - V) / 50 * 1000 * 50 / (V1 - V)$ mg of CaCO_3 equivalent
= $(V2 - V / V1 - V) * 1000$ mg of CaCO_3 equivalent
= **$(V2 - V / V1 - V) * 1000$ mg/L or ppm**

3) Determination of Permanent Hardness:

Since, 50 mL boiled water sample = $(V3 - V)$ mL Soap solution

Therefore, 1000 mL of boiled water sample = $(V3 - V) / 50 * 1000$ mL of Soap solution

from eq. 2

Permanent Hardness = $(V3 - V) / 50 * 1000 * 50 / (V2 - V)$ mg/L
= $(V3 - V / V2 - V) * 1000$ mg/L

4) Determination of Temporary Hardness

Temporary Hardness = **Total hardness - Permanent Hardness**

Alkalinity and its determination

1

"Alkalinity is a measure of the ability of a water sample to neutralize acids."

Sources of Alkalinity:

Natural sources: Dissolution of rocks and minerals (e.g., limestone containing CaCO_3), atmospheric CO_2 .

Anthropogenic sources: Industrial effluents, detergents, and water treatment chemicals.

Chemical species contributing to alkalinity

The major contributors are:

- Hydroxide ions (OH^-)
- Carbonate ions (CO_3^{2-})
- Bicarbonate ions (HCO_3^-)

Other weak acid anions (e.g., borates, silicates, phosphates, etc.) may contribute, but usually to a lesser extent.

Alkalinity and its determination

Principle of Alkalinity Determination: Alkalinity is determined by **acid-base titration** of a known volume of water sample using a **standard acid solution** and indicators to detect titration endpoints.

Phenolphthalein indicator

- ✓ Endpoint pH ≈ 8.3
- ✓ Pink to colorless
- ✓ Determines phenolphthalein alkalinity (P)

Methyl orange indicator

- ✓ Endpoint pH ≈ 4.5
- ✓ Yellow to orange
- ✓ Determines total alkalinity (T)

Alkalinity and its determination

Principle of Alkalinity Determination: Alkalinity is determined by **acid-base titration** of a known volume of water sample using a **standard acid solution** and indicators to detect titration endpoints.

Phenolphthalein Indicator (P)

- ✓ Endpoint pH $\approx$ 8.3
- ✓ Pink to colorless
- ✓ Determines phenolphthalein alkalinity (P)

Methyl Orange Indicator (M)

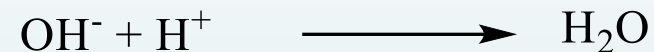
- ✓ Endpoint pH $\approx$ 4.5
- ✓ Yellow to orange
- ✓ Determines total alkalinity (T)

Alkalinity and its determination

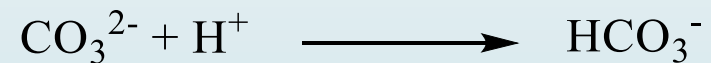
Titration Reactions

The **acid** reacts stepwise with the **alkaline** species in water:

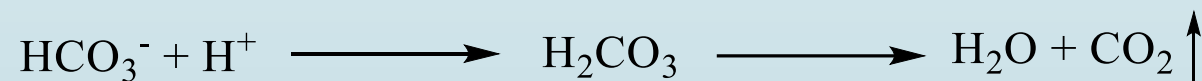
Neutralization of Hydroxide:



Conversion of **Carbonate** to **Bicarbonate**:



Conversion of Bicarbonate to **Carbonic Acid**:



P

M

Calculation: Based on the volume of acid used up to each endpoint

► Important Parameters and Limits:

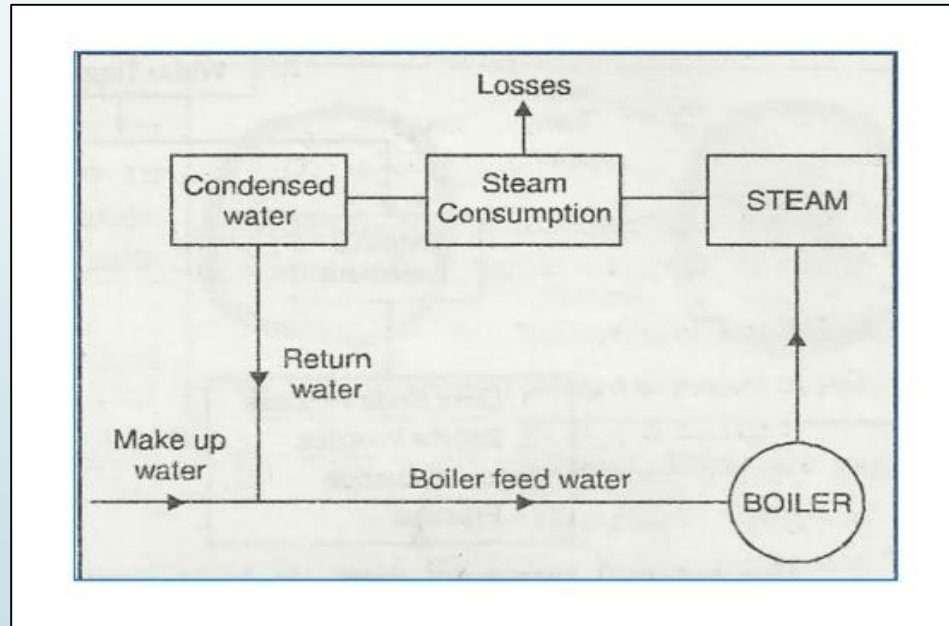
- pH: 6.5 – 8.5
- Hardness: 200 mg/L (desirable), 600 mg/L (maximum permissible)
- Chloride: 250 mg/L
- Fluoride: 1.0 mg/L
- Nitrate: 45 mg/L
- Total Dissolved Solids (TDS): 500 mg/L (desirable), 2000 mg/L (maximum)
- Iron: 0.3 mg/L
- Turbidity: 1 NTU (desirable), 5 NTU (maximum)
- Residual free chlorine: 0.2 mg/L (min after 30 min contact)

Remarks: If water doesn't meet desirable limits, alternate sources or treatment are advised.
The maximum permissible limits apply when no alternative is available.

BOILER

‘A boiler is a closed vessel in which water is heated to produce steam by applying heat energy.’

Purpose: To generate steam for power generation, heating, and food processing.



Boiler feed water

Water used in the boiler in this regard is termed as **boiler feed water**.

Boiler feed water = Return water + Makeup water

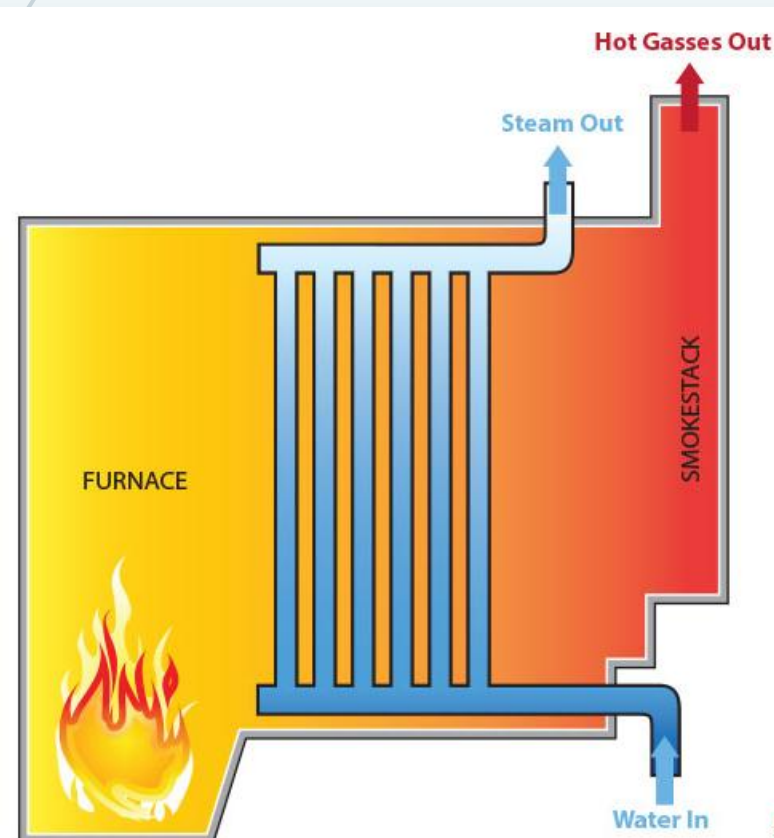
Type of boiler

7

Water tube boiler

Working Principle:

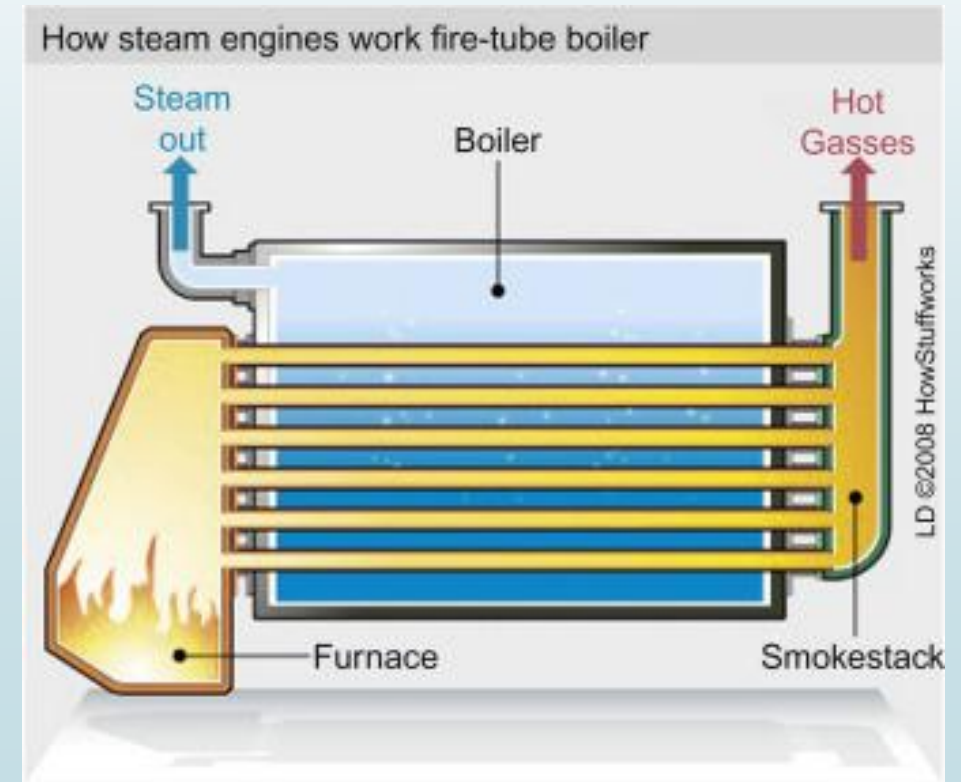
- Water circulates inside the tubes.
- Hot flue gases pass over the tubes.
- Heat is transferred to the water, producing steam.



Fire tube boiler

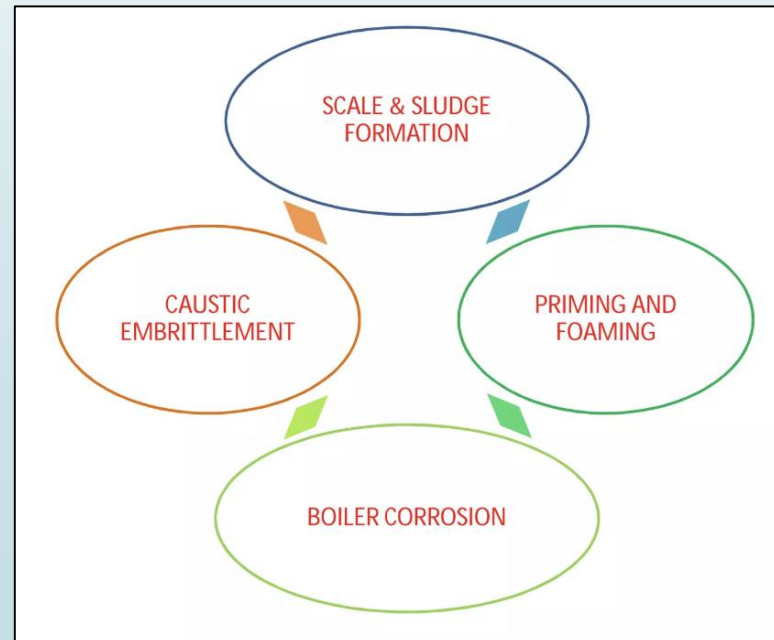
Working Principle:

- Hot flue gases flow inside the tubes,
- Transferring heat to the surrounding water, which converts into steam.



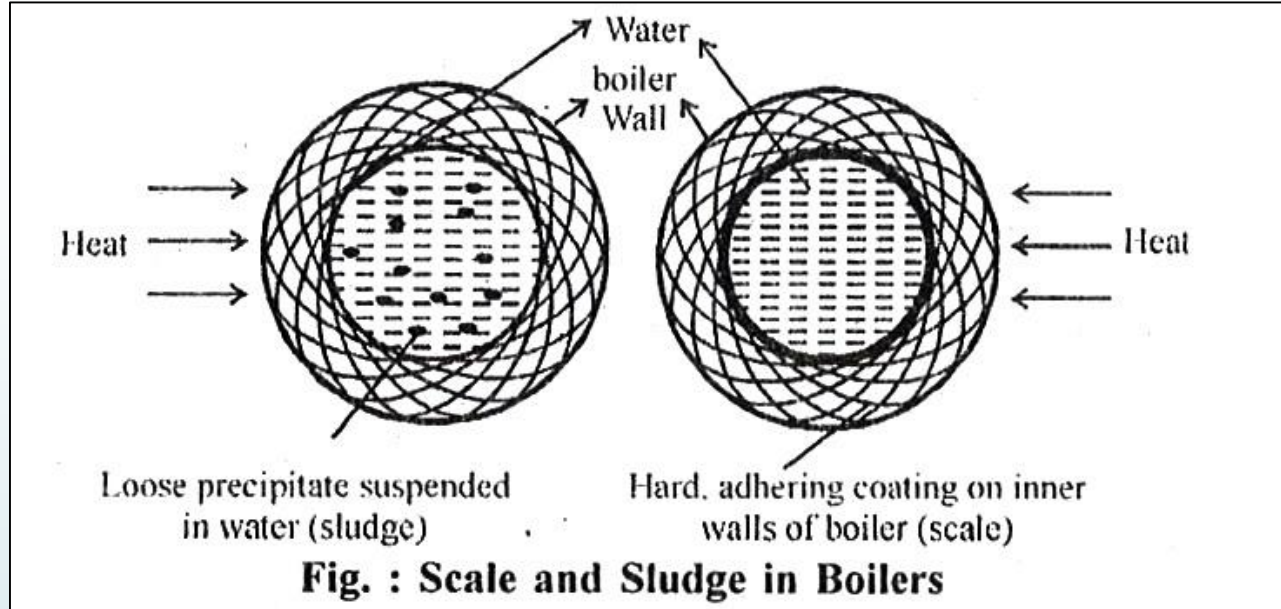
Boiler Problems/Troubles

- Sludge and scale formation
- Corrosion of Boiler metal
- Caustic Embrittlement
- Priming & Foaming



Sludge and Scale

10



Sludge & its Formation

- Soft, loose & slimy precipitate.
- Formed by substances which have greater solubility in hot water than in cold water
e.g., MgCO_3 , MgCl_2 , MgSO_4 , CaCl_2 , etc.
- Formed at a colder place of the boiler and get collected at place where the flow rate is slow.
- It can be easily removed with a wire brush.

Disadvantages of sludge formation:

- i) Poor conductor of heat, which decreases boiler efficiency.
- ii) Excessive sludge formation hampers the working of the boiler.
- iii) Choking of pipelines

Prevention of Sludge formation:

- i) By using softened water
- ii) By the frequent removal of concentrated water.

Scale & its Formation

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- Scale is a hard deposit sticks to the inner walls of the boiler.
- Scales are the main source of boiler trouble.
- Substances responsible- CaSO_4 , Ca(OH)_2 , CaCO_3 , $\text{Ca(PO}_4)_2$

Formation of scale:

- i) Decomposition of $\text{Ca(HCO}_3)_2$
- ii) Decomposition of CaSO_4
- iii) Hydrolysis of Magnesium Salts

Disadvantages of scale formation:

- i) Wastage of fuel
- ii) Lowering of the boiler's working pressure
- iii) Things about tube walls
- iv) Decrease in efficiency

Prevention of scale formation:

a) By external treatment-

- Removal of hardness
- Ion exchange process
- Zeolite method
- Lime-soda process

b) By internal treatment- addition of chemicals to remove scale-causing substances

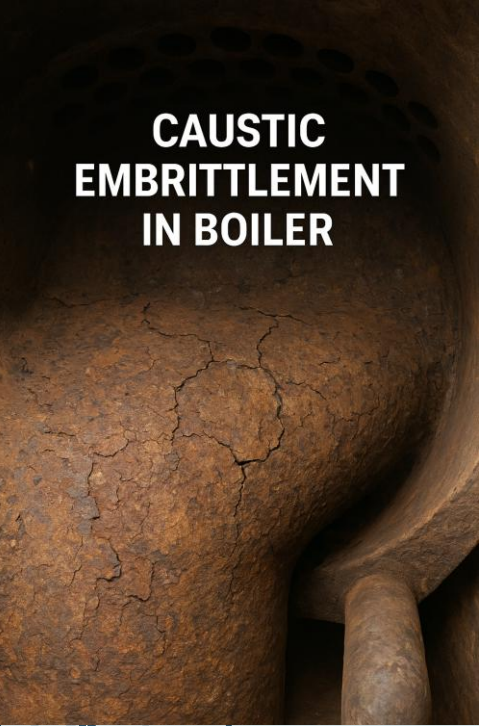
- Colloidal conditioning
- Carbonate conditioning
- Phosphate conditioning
- Calgon conditioning

Corrosion of boiler metal



- Corrosion is the gradual destruction of boiler metal due to chemical/electrochemical reactions with water and dissolved gases.
- Main corrosive agents: **dissolved O_2 , CO_2 , and acidic impurities.**
- **Oxygen corrosion** causes pitting and rust formation (Fe_2O_3), weakening metal parts.
- **CO_2 corrosion** forms carbonic acid, attacking condensate lines.
- **Acid corrosion** occurs at low pH or with acidic salts, leading to uniform metal thinning.
- Results: **metal failure**, leakage, lower efficiency, higher maintenance.
- Prevention: **deaeration, oxygen scavengers** (e.g., sodium sulfite, hydrazine), **pH control**, and **proper water treatment**.
- Regular **inspection and maintenance** detect and prevent severe damage.

Caustic embrittlement



CAUSTIC EMBRITTLEMENT IN BOILER

The phenomenon in which the material of a boiler becomes brittle due to the accumulation of caustic substances is known as **Caustic Embrittlement**.

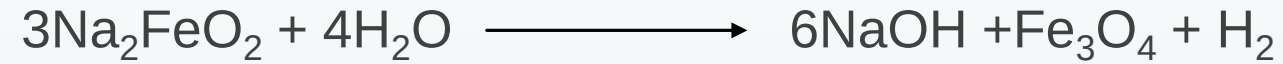
- As water evaporates in the boiler, conc. of Sodium carbonate increases.
- Sod. Carbonate is also used in water softening by lime-soda process due to this some sodium carbonate may be left behind.
- As the conc. of sod. carbonates increase, it undergoes hydrolysis to form sod. Hydroxide.



- Presence of NaOH makes the water alkaline.

**CAUSTIC
EMBRITTLMENT
IN BOILER**

- NaOH attacks the surrounding material & dissolves the iron of the boiler as Sodium ferrate($\text{Na}_2\text{Fe}_2\text{O}$).
- It decomposes, leading to the rusting of the boiler



- This causes embrittlement of boiler-like rivets, bends & joints.

Prevention:

- i. By using Sodium Phosphate instead of sodium carbonate as a softening reagent.
- ii. By adding tannin or lignin to boiler water, it blocks the hairline crack and prevents infiltration of caustic salts into these areas
- iii. Adding Na_2SO_4 to boiler water also blocks hairline cracks.

Priming and Foaming

17

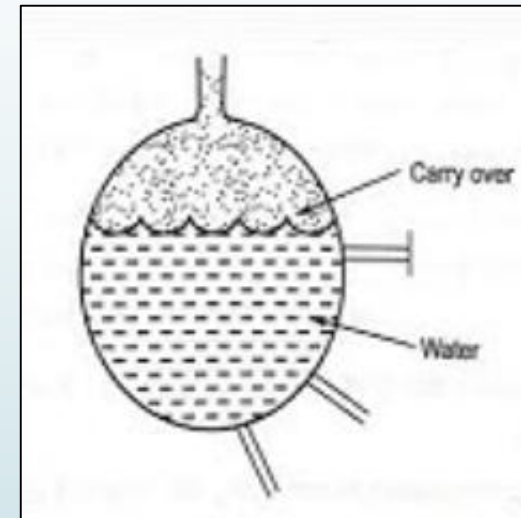
(a) Priming

When the steam is generated rapidly in the boilers, some droplets of liquid water are carried along with the steam. This problem is known as **Priming**.

Causes of priming:

- i) Due to the presence of suspended impurities & to some extent to dissolved impurities.
- ii) Due to algae & vegetables.
- iii) Due to the high water level in the boiler.
- iv) Due to the high velocity of steam, which carries water.

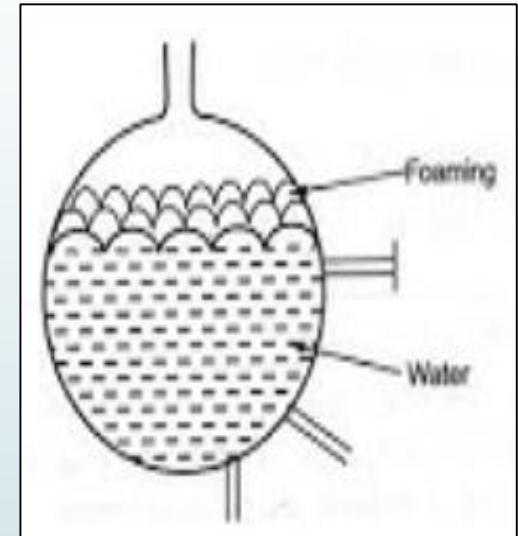
Prevention: By using soft water, fitting mechanical steam purifiers, controlling steam velocity, improving the design of the boiler, and removing scale & sludge on a regular basis.



PRIMING

(b) Foaming

- The formation of persistent foam or bubbles on the surface of boiler water, which do not break or dissolve easily, is known as **foaming**.
- **Causes:** The presence of dissolved substances such as oil, grease, and alkaline impurities leads to foaming. Oil and alkali react to form soap, which lowers the surface tension of water and increases foaming.
- **Prevention:**
 - i) By using antifoaming chemicals like castor oil.
 - ii) By adding coagulants such as aluminate.
 - iii) By removing foaming agents from water.
 - iv) By controlling the concentration of salts & sludge.



FOAMING

Specification of boiler feed water

Purpose:

- To prevent scale formation, corrosion, and boiler damage.
- To ensure efficient heat transfer and long boiler life.

It should be free from:

- ❖ **Hardness** (Sludge & scale formation)
- ❖ **Organic matter, oil, grease** (Foaming)
- ❖ **Dissolved gases like CO_2 , H_2S , O_2** (to avoid corrosion)
- ❖ **Alkalinity** (caustic embrittlement)
- ❖ **Turbidity and sediments**

Treatment of boiler feed water

There are two methods of treatment:

a) Internal treatment methods- (water treatment inside the boiler)

can be achieved by adding certain chemicals to the water in the boiler

- i) Phosphate conditioning
- ii) Calgon conditioning
- iii) Sodium meta-aluminate
- iv) Carbonate conditioning
- v) Colloidal conditioning

b) External treatment methods- (water treatment before being fed into boilers)

- i) Lime soda process
- ii) Zeolite Process
- ii) Ion exchange process

Internal treatment methods

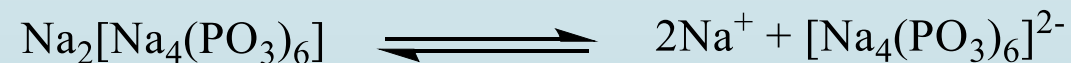
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► Phosphate conditioning:

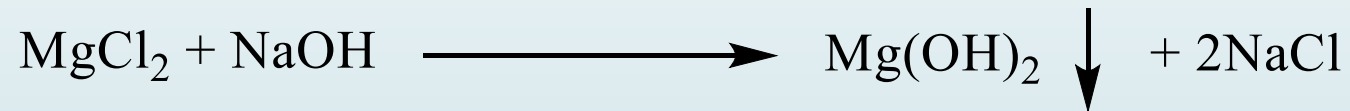
In high-pressure boilers, sodium phosphate is added to react with hard water salts, forming a soft and easily removable sludge of calcium and magnesium phosphates. This sludge is then removed through the blowdown operation, preventing scale formation.



► **Calgon conditioning:** Calgon (sodium hexametaphosphate, SHMP) is added to boiler water to prevent scale formation. It reacts with CaSO_4 to form a soluble complex compound, which remains dissolved in water and does not harm the boiler.



- **Sodium Meta Aluminate Treatment:** Sodium meta aluminate (NaAlO_2) is added to boiler water to **soften hard water** and **control sludge**. It undergoes **hydrolysis** to form sodium hydroxide and aluminum hydroxide:



Sodium aluminate, after hydrolysis, gives NaOH, which converts MgCl_2 into Mg(OH)_2 . Both Mg(OH)_2 & Al(OH)_3 can be removed by the blowdown process.

Carbonate Conditioning: In low-pressure boilers, scale formation can be avoided by adding **sodium carbonate (Na_2CO_3)** to the boiler water. The carbonate reacts with the hardness-causing salts to form **a precipitate**, which can then be removed through a **blowdown operation**.



Colloidal Conditioning: In low-pressure boilers, scale formation can be minimized by adding organic substances such as **kerosene**, **tannin**, or **agar-agar** gel to the boiler water. These substances form a **protective coating over the scale-forming precipitates**, keeping them in a finely divided (colloidal) form, which can then be **removed by blowdown operation**.

Treatment of boiler feed water

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- i) Lime soda process
- ii) Zeolite Process
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Internal treatment methods

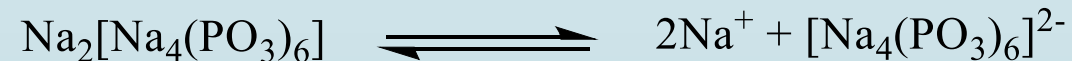
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► Phosphate conditioning:

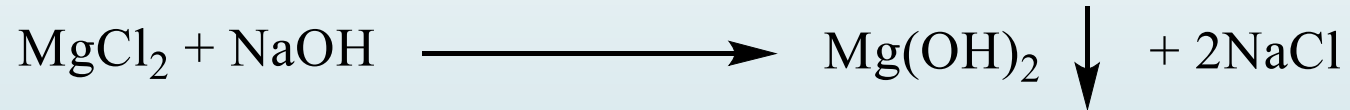
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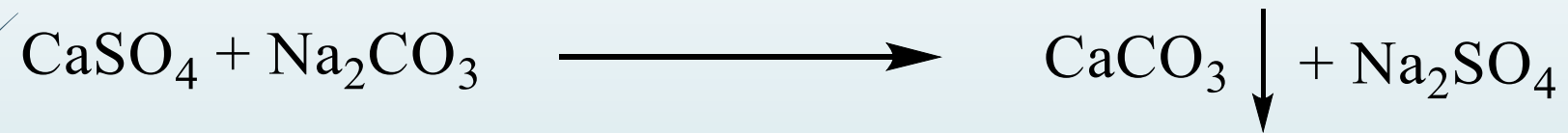


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External treatment methods (Water softening method)

- a) **External treatment methods**-(water treatment before being fed into boilers)
 - i) Lime soda process
 - ii) Zeolite Process
 - ii) Ion exchange process

Lime-soda process

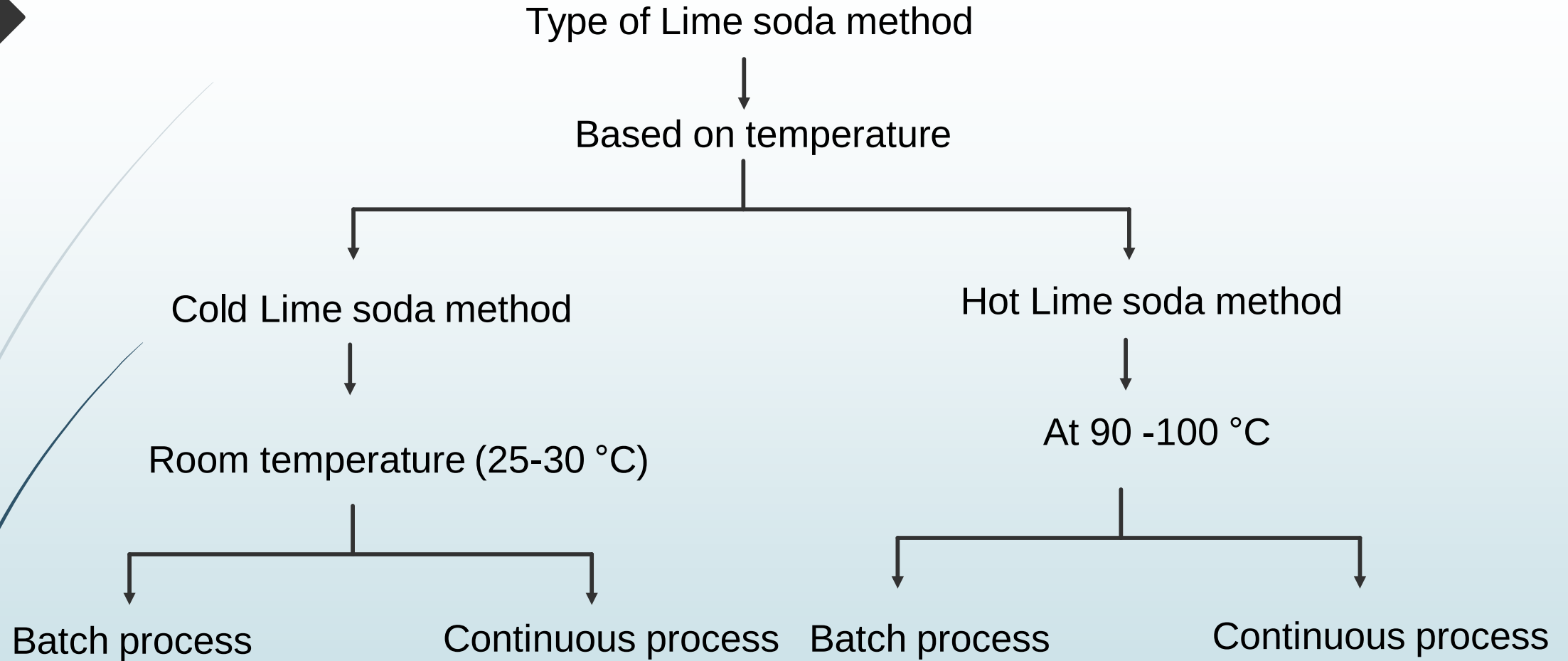
- Hard water contains dissolved **calcium** and **magnesium** salts that cause **scale formation** and **reduce soap efficiency**.
- Water softening is the process of **removing** hardness-producing salts.
- The Lime-Soda Process is one of the oldest and **most economical** methods for softening water.

Principle of Lime–Soda Process: In this method, **soluble hardness**-causing impurities are converted into **insoluble precipitates**, which can be removed by **settling** and **filtration**.

Soluble salts $\longrightarrow$ Insoluble precipitate

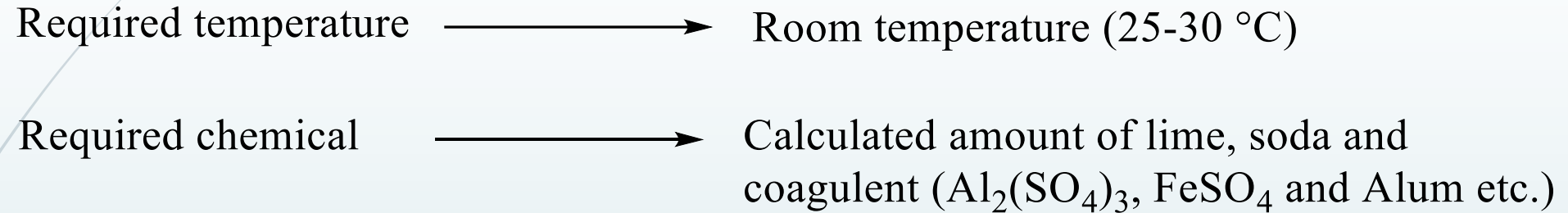
Lime $[\text{Ca}(\text{OH})_2]$ $\rightarrow$ removes **temporary** & part of **permanent** hardness.

Soda $[\text{Na}_2\text{CO}_3]$ $\rightarrow$ removes remaining **permanent** hardness.

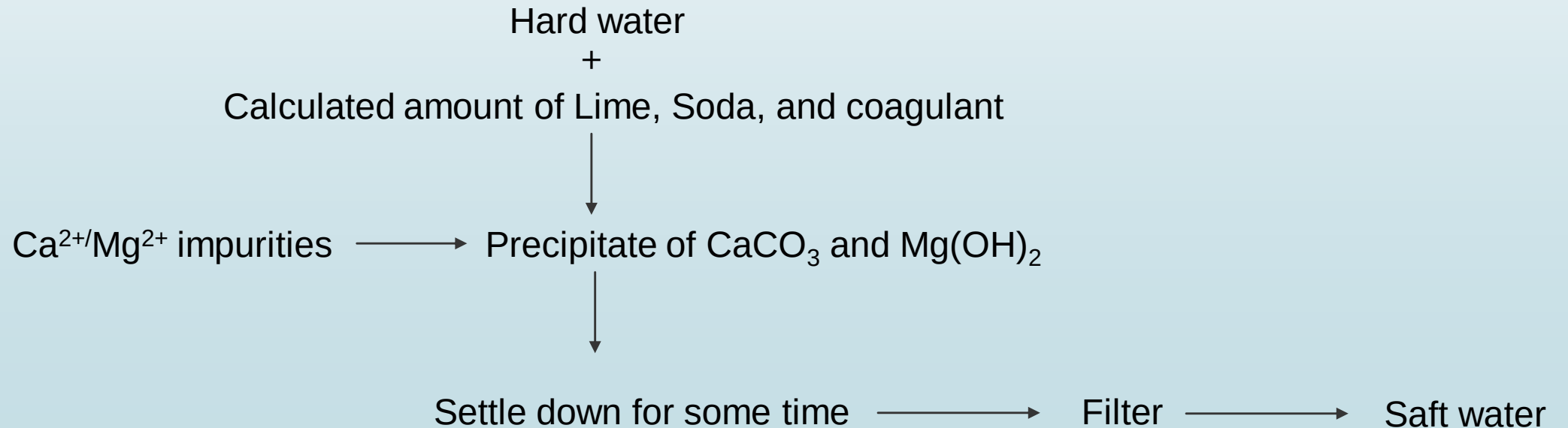


Types of lime-soda process

Cold Lime soda process:



Water softening by the cold lime soda process:



Cold Lime soda process:

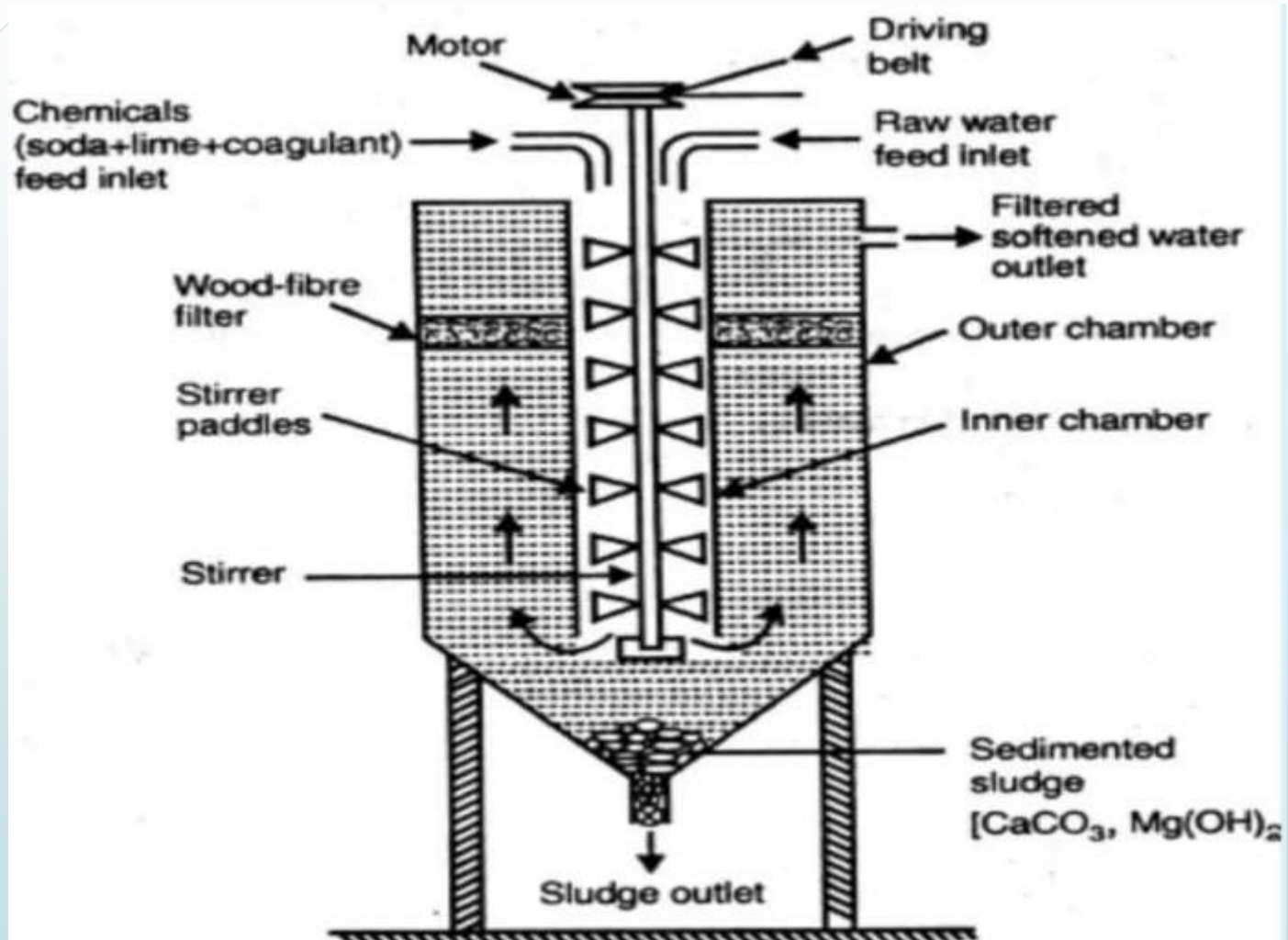


Figure 1. Basic unit used for Cold- Lime soda process

Hot- Lime soda process

10

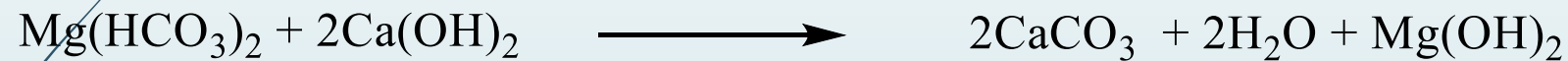
The hot-lime soda process is a water softening method carried out at **elevated temperatures (90–100 °C)** to remove hardness-causing salts.

- Operating Conditions:
 - Process carried out at 94-100 °C in very dilute solutions (~0.001 M).
 - High temperature increases reaction rate and **reduces water viscosity**.
 - Promotes faster aggregation and settling of precipitated particles.
- Coagulant:
 - **No coagulant is required**, unlike the cold process.
- Water Quality:
 - Produces soft water with residual hardness of 17-34 ppm.
- Efficiency:
 - **Highly efficient and suitable for industrial applications where high-quality soft water is required.**

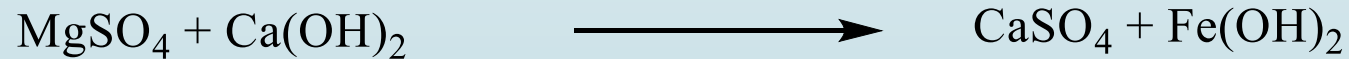
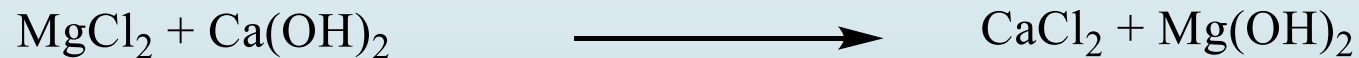
Chemical reaction involved in the lime soda method

12

Lime remove all temporary hardness



Lime removes dissolved magnesium, iron & aluminium salts



Chemical reaction involved in the lime soda method

13

Lime removes dissolved gases



Lime removes mineral acid



Chemical reaction involved in the lime soda method

14

Soda removes all permanent hardness



Advantages & disadvantages of lime soda process

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► Advantages-

- **Economical process:** Cost-effective compared to other softening methods.
- **Increases pH:** The rise in pH during treatment helps **reduce pipe corrosion** and **inhibits the growth of pathogenic bacteria**.
- **Reduces scale formation:** Effectively **removes temporary** and a significant portion of permanent hardness, minimizing scaling in boilers and pipes.
- **Improves water quality:** Makes water suitable for various domestic and industrial purposes.

► Disadvantages-

1. Excess sludge formation
2. Requires skilled supervision
3. Hardness up to 15 ppm can be only removed.

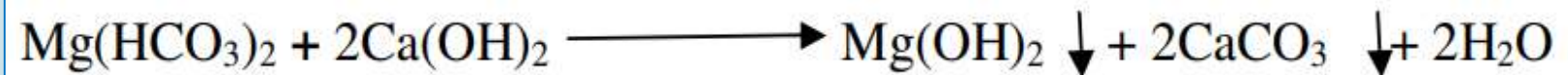
Key points to remember

Based on the various reactions discussed in the previous slide on the lime-soda process, the following deductions can be made:

- 1 equivalent of calcium temporary hardness requires 1 equivalent of lime for its removal.



- 1 equivalent of magnesium temporary hardness requires 2 equivalents of lime for its removal.



- 1 equivalent of calcium permanent hardness requires 1 equivalent of soda for its removal.



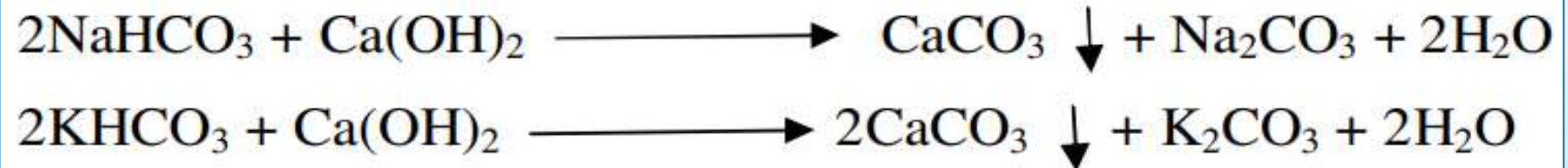
- 1 equivalent of magnesium permanent hardness requires 1 equivalent of lime and 1 equivalent of soda for its removal.



Lime reacts with HCl , H_2SO_4 , CO_2 , H_2S , and salts of iron, aluminium, etc., which can contribute to calcium permanent hardness. Therefore, their respective equivalents must be considered when calculating the lime requirement.



- The calcium permanent hardness generated during the process should also be considered when calculating the soda requirement.
- 2 equivalents of HCO_3^- require 2 equivalents of lime for removal.



Note: Substances like KCl , NaCl , SiO_2 , Na_2SO_4 , Fe_2O_3 , K_2SO_4 etc. do not contribute towards hardness. Therefore, It should be ignored while calculating the L-S requirement

Key Conversion:

- Suppose you have **100 parts of CaCO_3** hardness in water.
To remove it completely, you will need:
- **74 parts of lime $[\text{Ca}(\text{OH})_2]$**
- **106 parts of soda $[\text{Na}_2\text{CO}_3]$**

All hardness-causing substances should be converted into their corresponding CaCO_3 equivalents using the given formula.

$$\text{CaCO}_3 \text{ equiv.} = \text{Given wt. of impurity} * \frac{\text{Equiv. wt. of CaCO}_3}{\text{Equiv. wt. of impurity}}$$

Calculation

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Lime and Soda Calculation Formulae:

- 100 parts of CaCO_3 equivalent to:
 - 74 parts of Ca(OH)_2
 - 106 parts of Na_2CO_3

Lime requirement for softening:

$$\text{Lime Req.} = \frac{74}{100} \left\{ \begin{array}{l} \text{Temp. Ca}^{2+} + 2 * \text{Temp. Mg}^{2+} + \text{Perma (Mg}^{2+} + \text{Fe}^{2+} + \text{Al}^{3+}) + \\ \text{CO}_2 + \text{HCl} + \text{H}_2\text{SO}_4 + \text{HCO}_3^- + \text{NaHCO}_3 + \text{OH}^- - \text{NaAlO}_2 \end{array} \right\} * \text{Volume of water} \frac{* 100}{\% \text{ Purity}} \frac{* 1}{10^6} \text{ Kg}$$

All terms are expressed in CaCO_3 equivalents

Soda requirement for softening:

$$\text{Soda Req.} = \frac{106}{100} \left\{ \underbrace{\text{Perma (Ca}^{2+} + \text{Mg}^{2+} + \text{Fe}^{2+} + \text{Al}^{3+}) + \text{CO}_3^{2-} + \text{HCl} + \text{H}_2\text{SO}_4 + \text{OH}^- - \text{HCO}_3^-}_{\text{All terms are expressed in CaCO}_3 \text{ equivalents}} \right\} * \text{Valume of water} * \frac{100}{\% \text{ Purity}} * \frac{1}{10^6} \text{ Kg}$$

Temp. Ca^{2+} = $\text{Ca}(\text{HCO}_3)_2$, CaCO_3 , Temp. Ca^{2+}

Temp. Mg^{2+} = $\text{Mg}(\text{HCO}_3)_2$, MgCO_3 , Temp. Mg^{2+}

Perm. Ca^{2+} = CaCl_2 , CaSO_4 , $\text{Ca}(\text{NO}_3)_2$, Ca^{2+}

Perm. Mg^{2+} = MgCl_2 , MgSO_4 , $\text{Mg}(\text{NO}_3)_2$, Mg^{2+}

Perm. Fe^{2+} = FeSO_4 , Fe^{2+}

Numerical

Question: Calculate the amount of lime (84% pure) & soda (92% pure) required for treatment of 20,000 litres of water whose analysis is as follows

Substances	Given Weight (ppm)
$\text{Ca}(\text{HCO}_3)_2$	40.5
$\text{Mg}(\text{HCO}_3)_2$	36.5
CaSO_4	34.0
MgSO_4	30.0
CaCl_2	27.75
NaCl	10.0

Solution:

Substances	Given Wt. (ppm)	Multiplication Factor(M.F.)	CaCO ₃ Eq.(ppm) <i>=Given Wt.* M.F.</i>
Ca(HCO ₃) ₂	40.5	100/162	25
Mg(HCO ₃) ₂	36.5	100/146	25
CaSO ₄	34.0	100/120	25
MgSO ₄	30.0	100/136	25
CaCl ₂	27.75	100/111	25
NaCl	10.0	Ignored	It does not contribute to hardness

Hence, 84% pure lime is required for treating 20,000 litres of water

Lime Req.

$$= \frac{74}{100} \left\{ \begin{array}{l} \text{Temp. Ca}^{2+} + 2 * \text{Temp. Mg}^{2+} + \text{Perma (Mg}^{2+} + \text{Fe}^{2+} + \text{Al}^{3+}) \\ \text{CO}_2 + \text{HCl} + \text{H}_2\text{SO}_4 + \text{HCO}_3^- + \text{NaHCO}_3 + \text{OH}^- - \text{NaAlO}_2 \end{array} \right\} * \text{Valume of water} * \frac{100}{\% \text{ Purity}} * \frac{1}{10^6} \text{ Kg}$$

$$= \frac{74}{100} \left\{ 25 + 2*25 + 25 \right\} * 20,000 * \frac{100}{84} * \frac{1}{10^6} \text{ Kg}$$

$$= 1.76190 \text{ kg}$$

92 % pure soda is required for softening 20,000 litres of the water sample

$$\text{Soda Req.} = \frac{106}{100} \left\{ \begin{array}{l} \text{Perma (Ca}^{2+} + \text{Mg}^{2+} + \text{Fe}^{2+} + \text{Al}^{3+}) + \text{CO}_3^{2-} + \\ \text{HCl} + \text{H}_2\text{SO}_4 + \text{OH}^- - \text{HCO}_3^- \end{array} \right\} * \text{Valume of water} * \frac{100}{\% \text{ Purity}} * \frac{1}{10^6} \text{ Kg}$$

$$\text{Soda Req.} = \frac{106}{100} \left\{ 25 + 25 + 25 \right\} * 20,000 * \frac{100}{92} * \frac{1}{10^6} \text{ Kg}$$

$$= 1.728 \text{ kg}$$

Questions

1. Calculate the carbonate and non-carbonate hardness of a sample of water containing the dissolved salts as given below in mg/L. $\text{Mg}(\text{HCO}_3)_2 = 7.3$; $\text{Ca}(\text{HCO}_3)_2 = 40.5$; $\text{CaSO}_4 = 13.6$; $\text{MgCl}_2 = 21.75$ and $\text{NaCl} = 50$.
2. A sample of water is found to contain the following analytical data in mg/L. $\text{Mg}(\text{HCO}_3)_2 = 14.6$; $\text{MgCl}_2 = 9.5$; $\text{MgSO}_4 = 6.0$; $\text{Ca}(\text{HCO}_3)_2 = 16.2$. Calculate the Temporary and Permanent hardness of the sample of water.(Give answer in °F & °C units).
3. Calculate the amount of lime required for the softening of 10,000 Litres of hard water containing $\text{Mg}(\text{HCO}_3)_2 = 144$ ppm; $\text{MgCl}_2 = 95$ ppm; $\text{Ca}(\text{HCO}_3)_2 = 25$ ppm; $\text{CaCl}_2 = 111$ ppm; $\text{Fe}_2\text{O}_3 = 25$ ppm; $\text{Na}_2\text{SO}_4 = 15$ ppm.

UNIT 1

LECTURE- VI

Water softening process

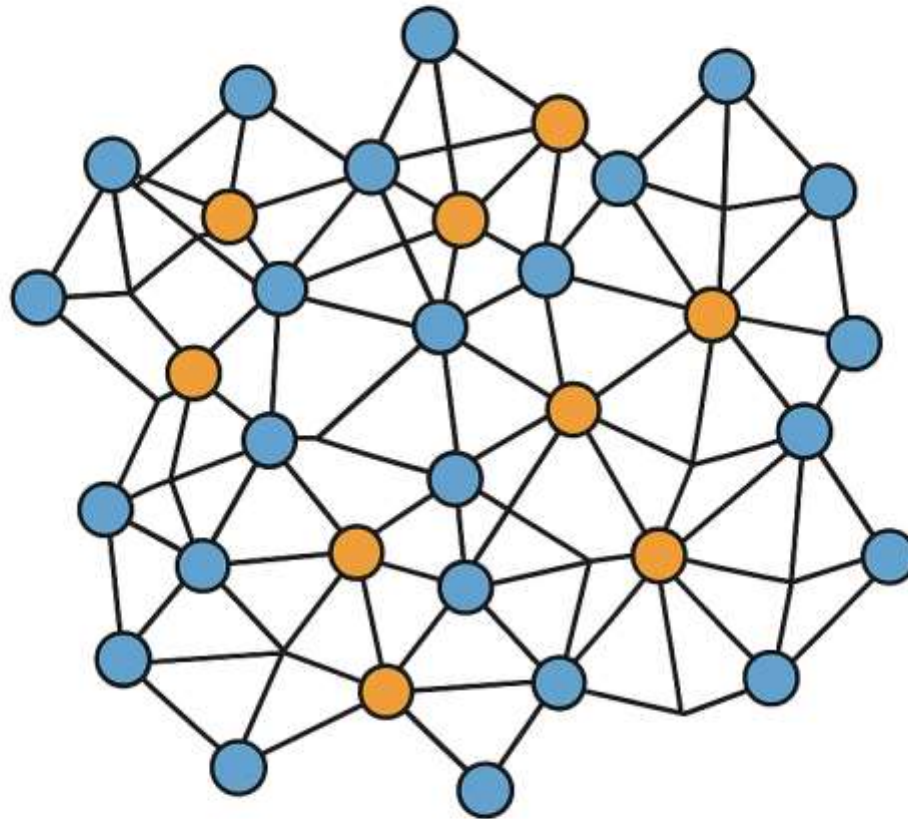
- 1. Zeolite Method**
- 2. Ion Exchange Method**

ZEOLITE METHOD

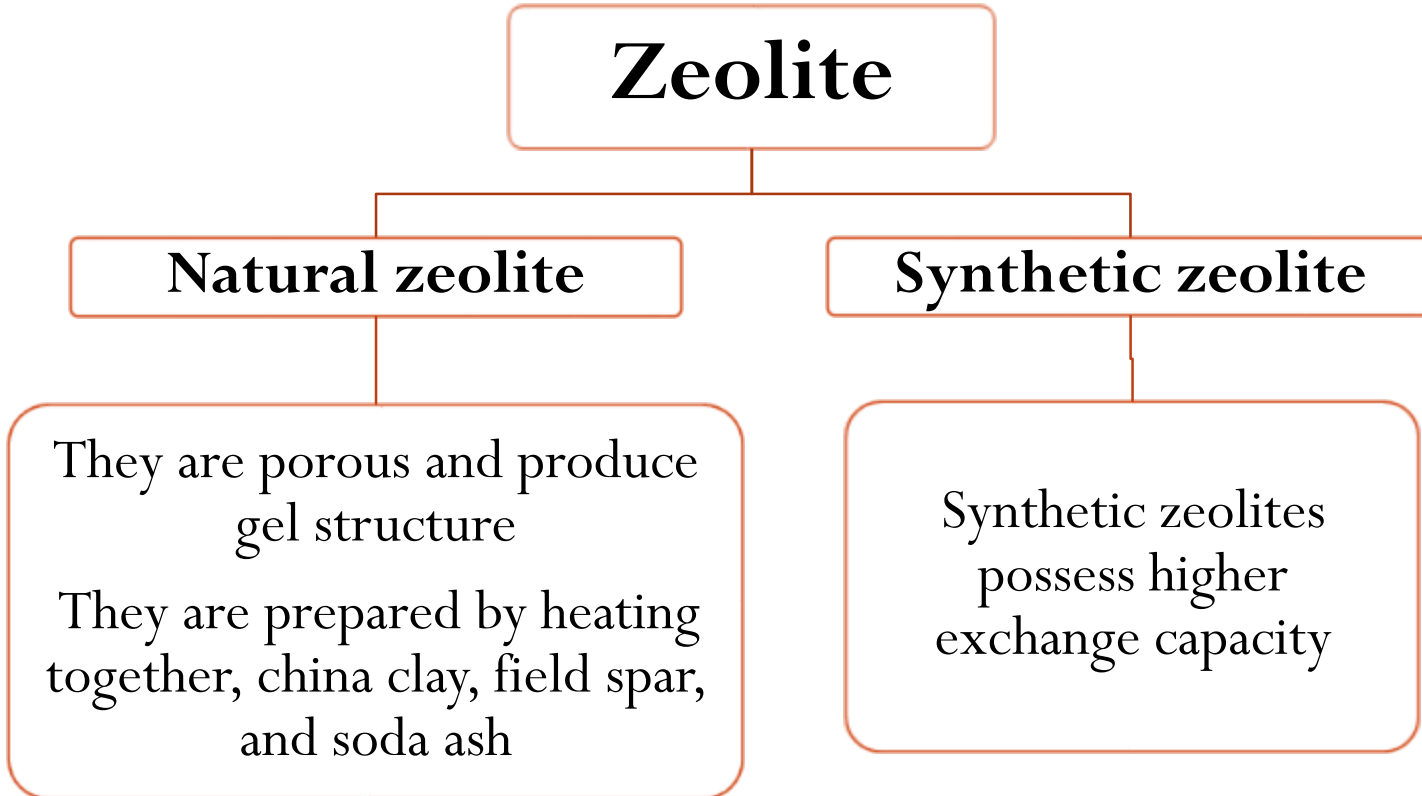
What is Zeolite?

- It is a hydrated sodium alumino silicate.
- It is represented as 'Ze'.
- It is capable of exchanging hardness, causing Ca^{2+} & Mg^{2+} .
- They have loosely held Na^+ ions.
- It is a white coloured powder.
- Zeolites are present naturally inside the earth.
- Chemical formula $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$, i.e. Na_2Ze , where x varies from 2-10 and y varies from 2-6.

STRUCTURE OF ZEOLITE



Types of Zeolite

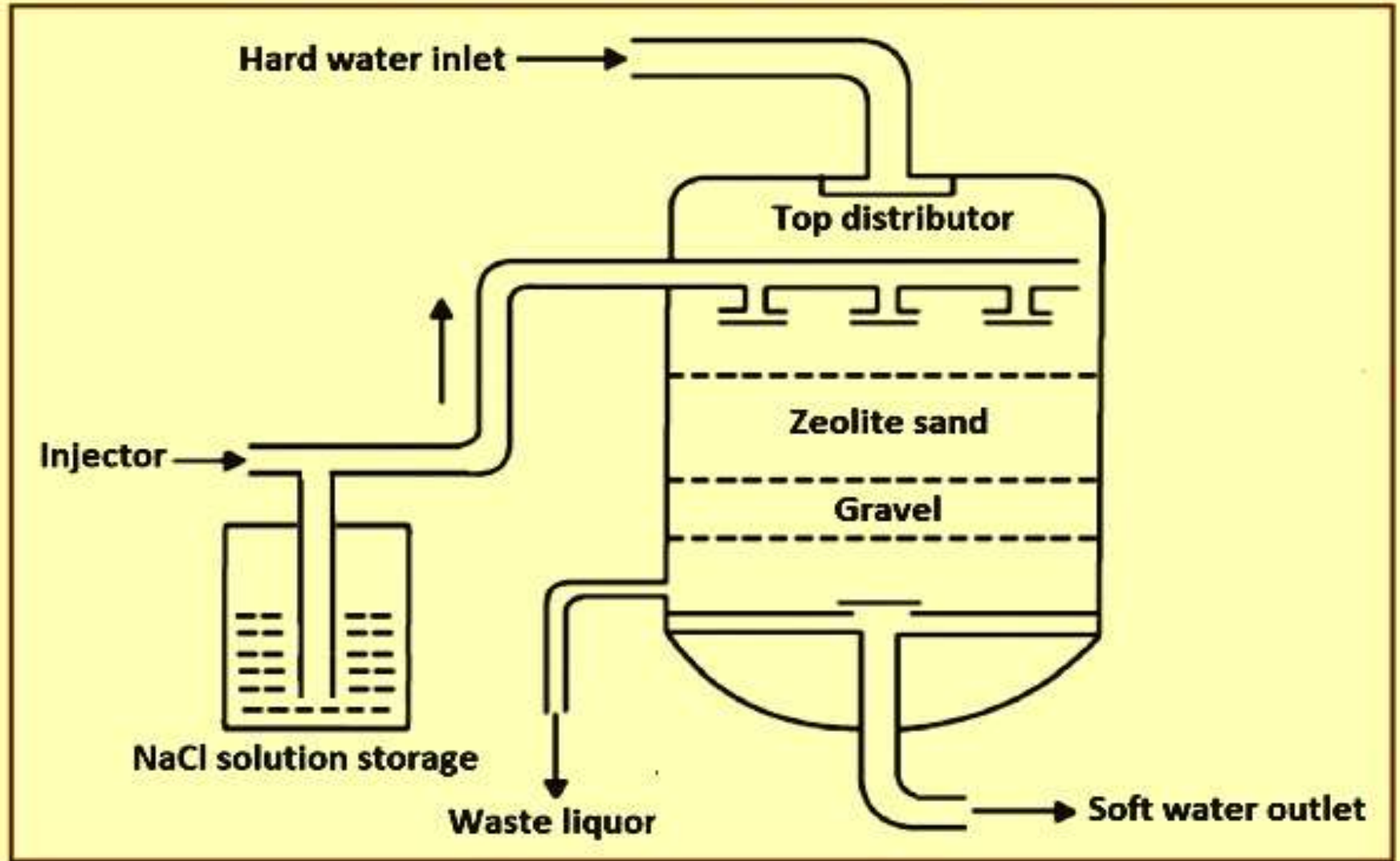


Principle

- When hard water containing heavy metal salts like Ca^{2+} and Mg^{2+} is passed through the sodium zeolite bed.
- The zeolite bed captures these heavy metal cations, and Na^+ ions are released.
- In this way, hard water is converted to soft water.

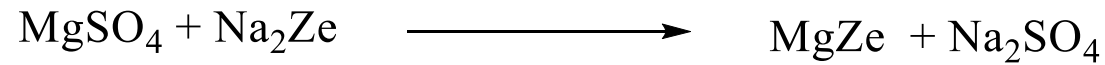
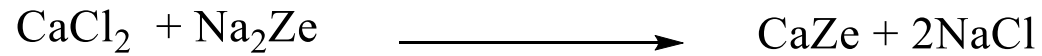
Process

Softening of hard water by Zeolite process



Reactions

During Treatment / Softening-



Regeneration

- After exhaustion of the Zeolite bed, it needs to be regenerated in order to keep filtering soft water from hard water
- For that, NaCl solution is sprinkled on the bed so that the Mg and Ca ions get replaced again with Na ions
- The washings are then removed.

During Regeneration

- $2\text{NaCl} + \text{CaZe} \longrightarrow \text{Na}_2\text{Ze} + \text{CaCl}_2$
- $2\text{NaCl} + \text{MgZe} \longrightarrow \text{Na}_2\text{Ze} + \text{MgCl}_2$

Advantages

- Simple to use
- Applicable for any hardness of water
- Activity can be easily checked by soap
- Equipment is compact and occupies less space.
- Requires less time for softening of water.

Limitations

- Cations like Fe^{2+} , Mn^{2+} are permanently exhausted by the Zeolite bed. Hence, their zeolite can't be easily regenerated.
- Warm water cannot be softened by the zeolite method because zeolite is soluble in warm water.
- Output water contains an equivalent amount of sodium salts.
- It removes only cations, not anions.
- Can not be used for turbid water.

Applications

- Used for softening of water.
- Used to extract precious metals.
- It is used as a catalyst.
- Used as an adsorbent.

ION EXCHANGE METHOD

- It is also called Deionization or Demineralization process.
- Demineralization of water is carried out by ion exchange resin.

Ion Exchange Resins

- Ion exchange resins are cross-linked structure, long chain organic polymer with a micro -porous structure.
- The functional groups attached to the polymeric chain are responsible for ion exchange.

Types of Resins

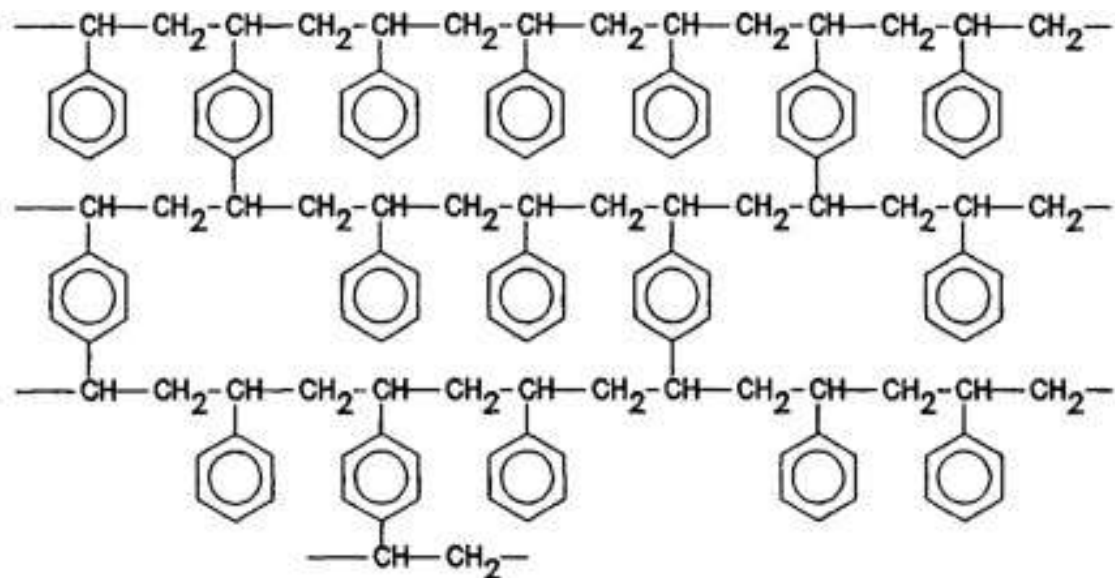
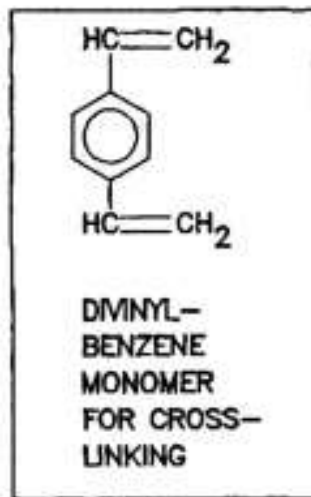
Cation Exchange Resin

- Resins containing acidic group like —COOH , $\text{—SO}_3\text{H}$ etc., are known as cation exchange resin.
- These are capable of exchanging their H^+ ions with cations of the solution.

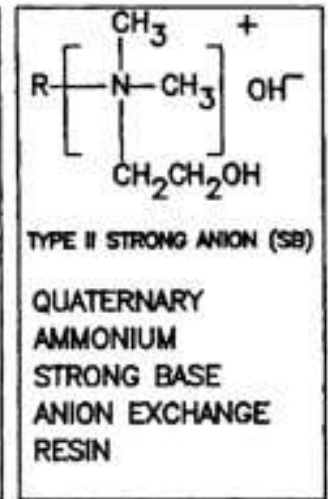
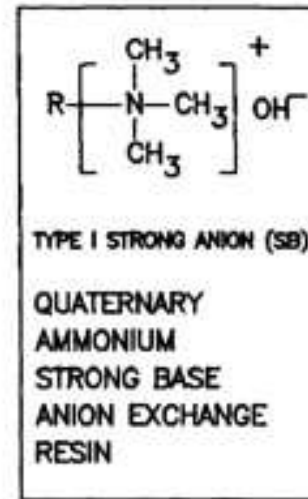
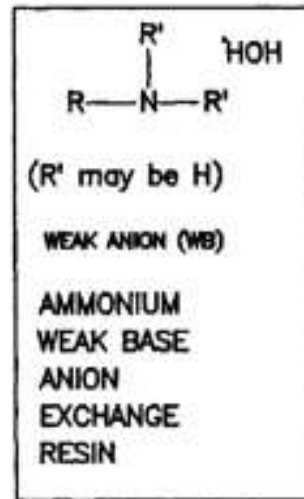
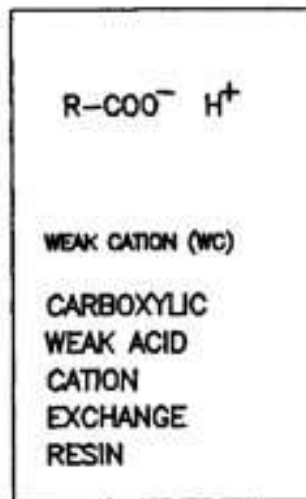
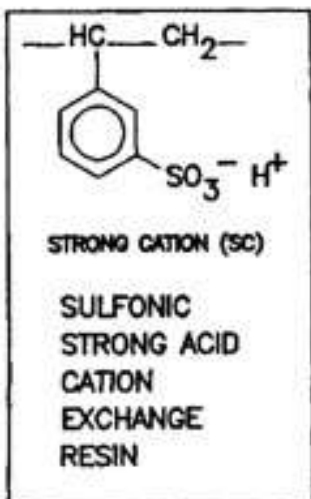
Anion Exchange Resin

- Resins having basic functional groups like Amine, substituted amine, etc., are known as anion exchange resins.
- These are capable of exchanging their OH^- ions or anions with anions of the solution.

Structure of Resins



POLYSTYRENE DMVINYL BENZENE BACKBONE



Process

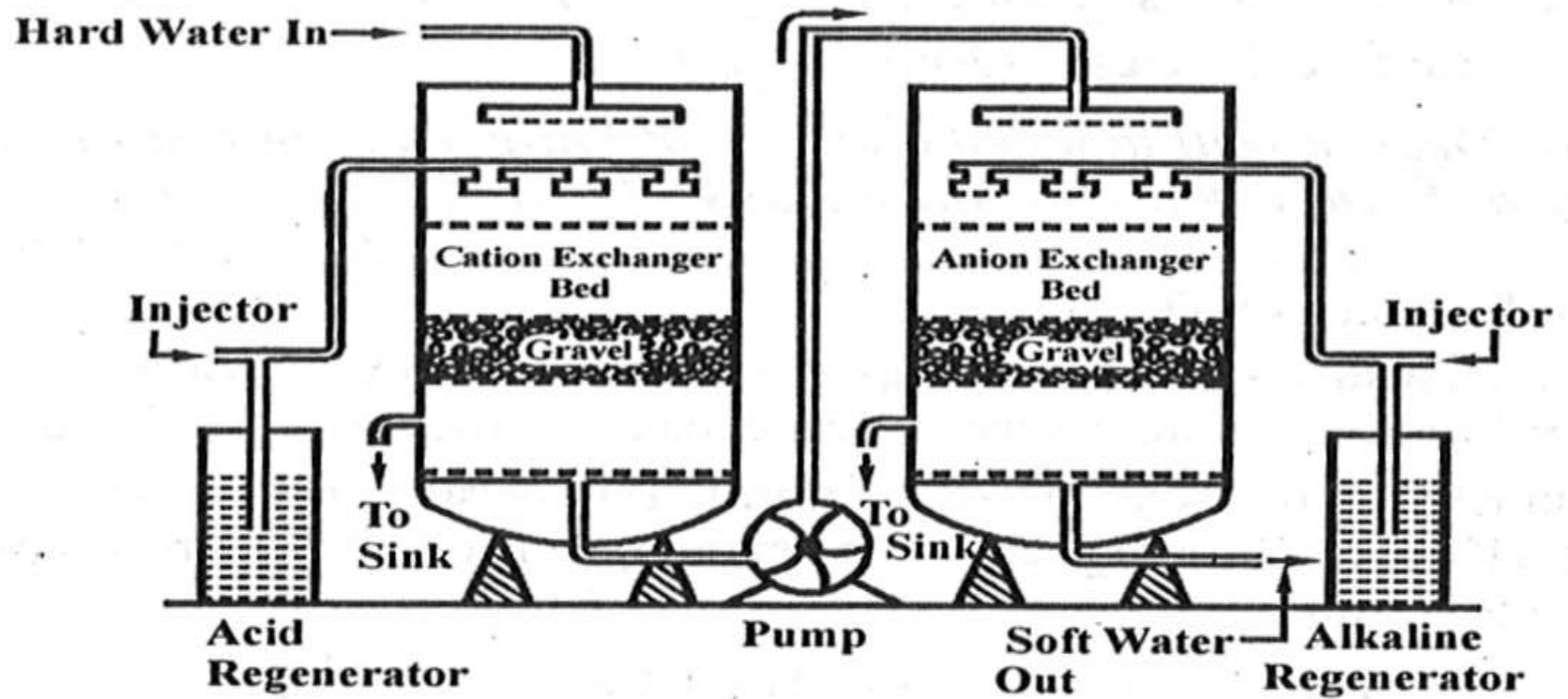
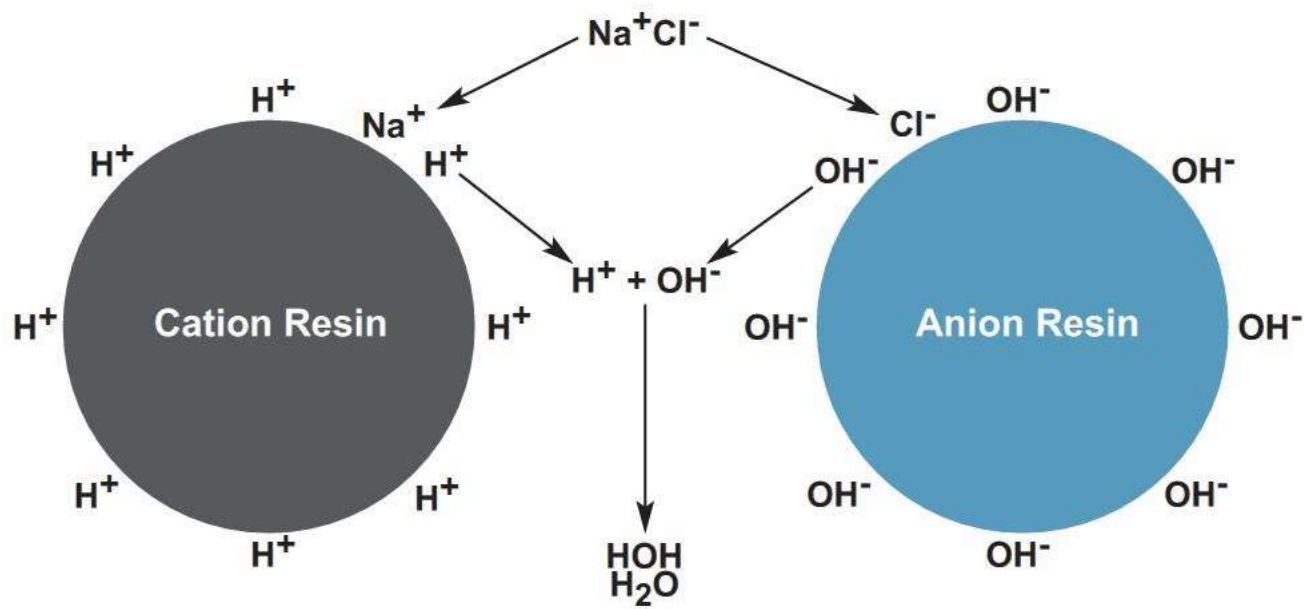
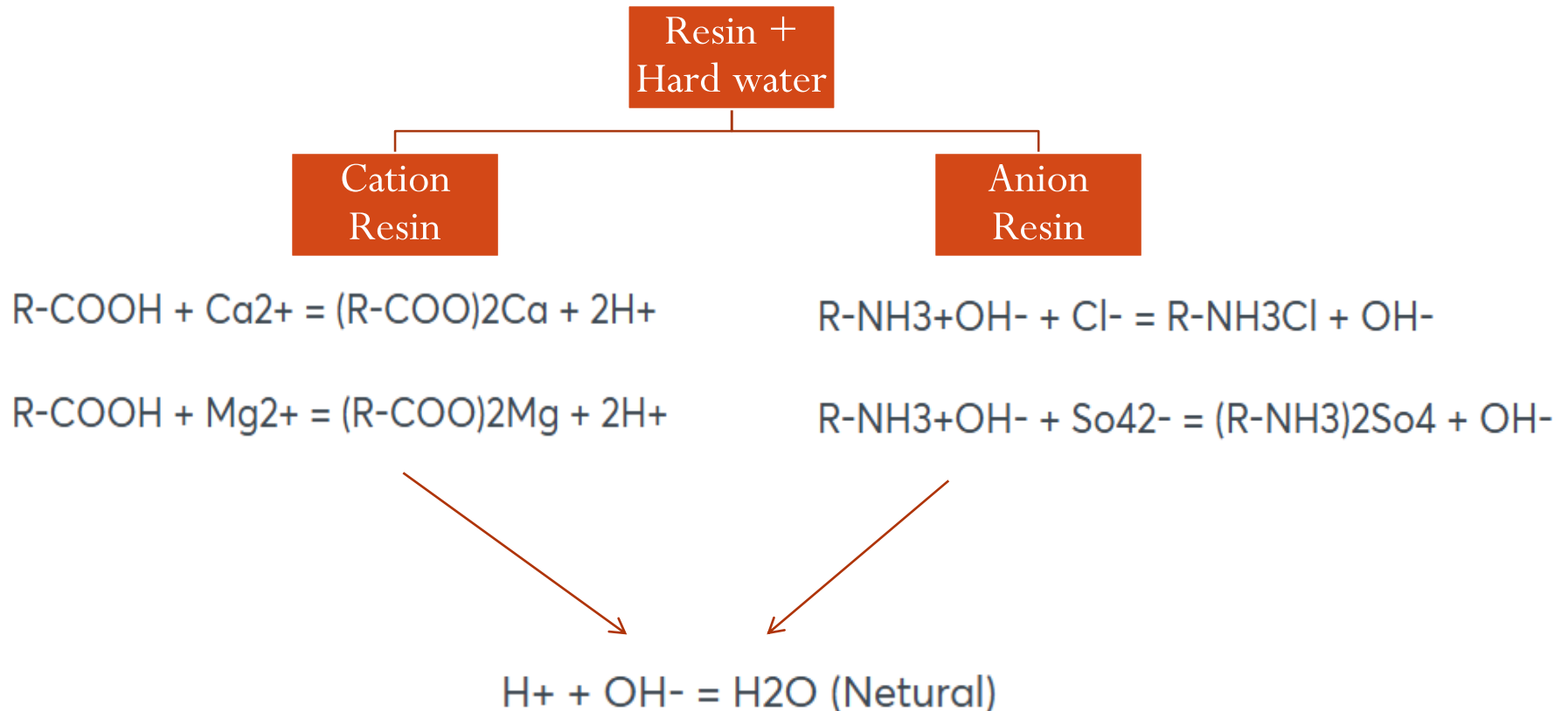


Fig. Demineralization of Water

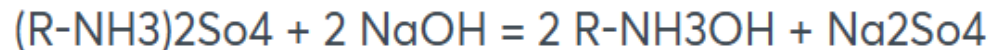
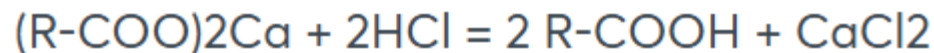


Reactions



Regeneration of resins

- When cation and anion exchangers are exhausted, regeneration should be done. The exhausted cation exchange resin is regenerated by reaction with dilute hydrochloric acid.



- The exhausted anion exchange resin is regenerated by reaction with dilute NaOH.

Mixed-bed Ion Exchange process

A single column containing cation and anion exchanger resins are taken in this process. When hard water is passed through a mixed –bed deionizer, it alternately comes in contact with anion exchangers and cation exchangers a number of times. So more efficient softening is done, and residual hardness of softened water is 0 to 1 ppm only. When the resin is exhausted, the bed is back washed. Due to its lower density, the anion exchanger forms an upper layer over the cation exchanger. The anion exchanger is regenerated by NaOH solution, and the cation exchanger is regenerated by sulphuric acid solution.

Advantages

- Can be used for highly acid and highly alkaline water.
- Residual hardness of water is as low as 2 ppm.
- Very good for treating water for high pressure boilers.

Disadvantages

- Expensive equipment and chemicals.
- Turbidity of water should be < 10 ppm.
- Otherwise output will reduce; turbidity needs to be coagulated before treatment.
- Needs skilled labour.

LIME SODA VS ZEOLITE VS ION EXCHANGE PROCESS

Lime-Soda

- It involves precipitation of Ca & Mg ions by adding Lime-Soda
- Produces sludge or precipitates that need proper disposal
- Regeneration is not applicable as the process involves the formation of insoluble precipitates, and the by product are removed from the system
- It produces 17-34 ppm (hot L-S) & 50-60 ppm (cold L-S) softened water.

Zeolite

- It involves passing water through a bed of zeolite, which exchange sodium ions for Ca & Mg ions
- Generate brine waste during regeneration which needs appropriate handling
- Requires regeneration by backwashing the zeolite bed with a brine solution
- It produces 10-15 ppm softened water.

Ion Exchange

- It involves the ion exchange between resins and Ca/Mg ions
- Generate brine waste during regeneration which needs appropriate handling
- Requires regeneration by passing a Acidic and Basic solution through Cation & Anion exchange bed respectively.
- It produces less than 2 ppm softened water.

Section

(A)

mcq -

(05)

Numore. -

10 + 5

theory -

15



(B)

(1)

mcq -

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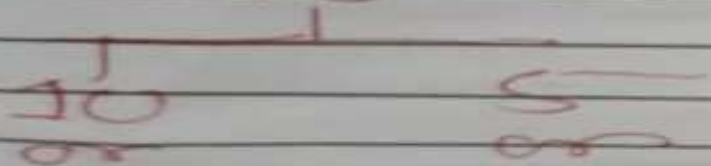
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Teacher's Signature